



Interfaces with Other Disciplines

Incentive schemes for resolving Parkinson's Law in project management

Bo Chen^{a,*}, Nicholas G. Hall^b^a Warwick Business School, University of Warwick, Coventry CV4 7AL, UK^b Fisher College of Business, The Ohio State University, Columbus, OH 43210-1144, USA

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ABSTRACT

Project management is a business process that supports about 30% of the world's economic activity. Yet projects routinely suffer from the influence of Parkinson's Law. This behavioural phenomenon routinely results in failure to deliver work that is completed early before its assigned deadline. As a consequence, the late completion of other work is not offset, and overall project performance suffers. Hence, project success rates below 40% are widely reported.

Our work uses mechanism design within non-cooperative game theory. A particular issue in the design process is to eliminate the possibility that a project worker with multiple dependent tasks can improve their incentive payment by falsely reporting some of their task completion times. From our review of the academic and business literature of project management, no incentive scheme used in practice accomplishes this.

Our results include the design of incentive schemes that eliminate or mitigate Parkinson's Law. These schemes apply to projects designed under either traditional Critical Path Method (CPM) planning or modern Critical Chain Project Management (CCPM) planning, and are also invulnerable to group strategy. A large-scale computational study validates the resulting benefit to project performance as substantial and also robust across different project characteristics. We also provide what is apparently the first analytical comparison between traditional CPM and modern CCPM planning systems.

The incentive schemes we propose are simple and easily implementable. We recognize that performance incentives are structured differently by each organization, but our work provides a flexible basis from which various practical schemes can be designed.

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1. Introduction

A project is a “temporary endeavor undertaken to create a unique product or service” (Project Management Institute 2013). As much as 30% of the world's economic activity is organized as projects (Hu, Cui, & Demeulemeester, 2015), implying an annual value of around \$27 trillion. Much of this activity is due to modern applications of project management, for example, information technology systems implementations, software development, and research and development. Further, shorter life cycles for products and services (Growwire.com, 2019) are motivating more frequent application of project management in the development of new products and services. Finally, more rapid corporate change is occurring, due to new technologies, more intense competition, and more dynamic customer preferences (Enlume, 2018), and the re-

sulting change requires professional project management. Reviews of the project management literature and discussions of research agendas appear in Hall (2012, 2013, 2016).

In a typical project, many task times are stochastic. Problematically, task times that are longer than expected typically increase project completion time. Whereas, task times that are shorter than expected fail to reduce project completion time. The primary reason for this asymmetry is *Parkinson's Law* (Parkinson, 1958), or “Work expands so as to fill the time available for its completion.” (Parkinson, 1955). As applied to project management, Parkinson's Law implies that the time which it takes to complete a task and report its completion is not less than the amount of time that is made available for it (Gutierrez & Kouvelis, 1991). The effect of Parkinson's Law is that the potential benefit of earlier than expected task completion is lost. This fails to compensate for the later than expected completion of other tasks. To resolve this issue, we describe market mechanisms that incentivize employee task

* Corresponding author.

E-mail addresses: b.chen@warwick.ac.uk (B. Chen), hall.33@osu.edu (N.G. Hall).

operators to report their early task completion times and thereby mitigate the effect of Parkinson's Law.

Parkinson's Law is well documented in standard project management texts (Klasterin, 2012; Wysocki, 2009). The project management literature includes many studies of Parkinson's Law. Krakowski (1974) provides several intuitive motivations for task operators to follow Parkinson's Law. First, there is a desire to appear busy, in order not to be allocated additional tasks. Quigley (2011) reports: "My grandfather worked for Rolls Royce during WW2, as a general laborer. The workforce did not trust management, such that if they worked harder in a given year and were rewarded with a bonus then they expected their quotas would be increased in the following years." The analogous concern in project management is that, if a task is completed early, less time will be allowed for it in future. Second, there is a desire to avoid having resources withdrawn, which can occur as a result of over-performance. Third, there is a natural tendency towards laziness or procrastination. Other factors motivating Parkinson's Law include a tendency to overestimate the importance of the task, or to obsess unnecessarily over small quality details.

We consider task operators who are internal to the company. Our study is concerned with seeking a solution to the behavioral issues of Parkinson's Law without convoluting other issues such as resource allocation. More specifically, our research objective is essentially to resolve *ceteris paribus* two behavioral issues, effort level and reporting, as influenced by Parkinson's Law. For example, consider a task with a deadline of ten days, where after six days the TO realizes that it is possible to complete the task in two more days, i.e., two days early. One possibility under Parkinson's Law is that the TO continues to work at full speed, completes the task after eight days, but fails to report completion to the PM until the deadline. A second possibility under Parkinson's Law, which we believe in practice is more prevalent is that the TO works more slowly over the next four days, and then completes the task at the deadline. However, from the perspective of the PM, these two behavioral models are operationally equivalent. Thus, our work implicitly models both reporting and effort level issues under Parkinson's Law.

A related behavioral phenomenon that adversely affects project performance is *Student Syndrome* (Klasterin, 2012). In Student Syndrome, a task operator delays the start of a task until the latest time at which, if later tasks match their expected durations, the project will complete without delay. However, where the expected duration of the task is exceeded, the project is delayed.

Traditional project planning is based on the critical path method (CPM) which was developed at DuPont Corporation for projects with deterministic task times, and PERT which was developed by Booz Allen Hamilton and the U.S. Navy for projects with stochastic task times. Schonberger (1981) discusses the issue of path merging, which results in PERT estimates being systematically too optimistic. Williams (1995) suggests that, the faster progress is made on a task, the greater is the degree of slowing down due to Parkinson's Law.

Gutierrez and Kouvelis (1991) model Parkinson's Law in project management, by using a work expansion factor. This factor represents "leisure time or unnecessary expansion introduced". This is consistent with our perspective that the delayed task completion and reporting times which occur under Parkinson's Law do not result from resource limitations. Consequently, they cannot be resolved by expediting, or "crashing" the tasks. In fact, when Parkinson's Law is in effect, crashing is likely to waste resources without improving the task time or project schedule. Consequently, we henceforth do not consider crashing. The authors develop a stochastic task completion time model, and use it to derive several implications for managerial decisions about the setting of task deadlines. Wu, Krishnan, and Ramachandran (2014) cite Parkin-

son's Law as a motivation for procrastination, resulting in distortion of workers' effort over time.

Goldratt (1997) proposes an alternative project management planning approach, critical chain project management (CCPM), which attempts to resolve the problem of Parkinson's Law. This is achieved in part by the removal of task deadlines from the project planning system, thereby eliminating expansion to fill available time. Steyn (2001) provides a supportive discussion of CCPM, which focuses on the use of buffer management to monitor and control the project. Despite numerous success stories in support of CCPM, its mechanism for alleviating Parkinson's Law is poorly supported. For example, Raz, Barnes, and Dvir (2003) comment, "Will they [the task operators] agree to shorten their duration estimates?...the behavioral aspects of identifying the precise amount of safety margin and taking it away from the task operators are dealt with only superficially...." A further concern about CCPM is the relative lack of project progress control that arises from the absence of task deadlines. An important question that remains is whether the effect of Parkinson's Law can be mitigated or eliminated entirely without removing the control mechanism of task deadlines. Our work designs incentive mechanisms to resolve this question.

A comprehensive introduction to mechanism design theory is provided by The Royal Swedish Academy of Sciences (2007). The study of market mechanisms originates from the work of Hurwicz (1960). A mechanism is *incentive compatible* (IC) if it specifies each agent's optimal choice, irrespective of what other agents do, i.e., it implies a dominant strategy equilibrium. Myerson (1979, 1982, 1986) shows that any equilibrium outcome of an arbitrary mechanism can be replicated by an IC direct mechanism, a result that is known as the revelation principle. This result simplifies the search for an efficient mechanism.

Sommer and Loch (2009) develop a principal-agent model, where the principal is a senior manager and the single agent is an external project manager whose actions cannot be monitored by the principal. They examine the performance of adjustable incentive contracts under unforeseeable influences. By contrast, our work considers any number of agents who are internal to the project company. Kwon, Lippman, and Tang (2010) study a project-based system that consists of a project manager and a single contractor. They show that incentive contracts based on time, and some incentive contracts based on a fixed price, can coordinate the system; whereas, contracts based on a fixed price alone cannot do so. By contrast, our work considers multiple task operators within the project company. Kwon, Lippman, McCardle, and Tang (2012) consider an assembly-type project with a manufacturer and $n \geq 2$ suppliers with one task each. The tasks are identical, have no precedence requirements, and earn the same fixed payment. They compare a traditional payment scheme, under which each supplier receives payment on completion of his task, with a delayed payment scheme, under which all payments are made only on completion of the last task. Chen, Klasterin, and Wagner (2015) consider a similar problem, but with nonidentical tasks and a serial precedence network. They show that a contract containing a convex time-cost tradeoff dominates a fixed price contract, as measured by project schedule and profit. By contrast, the projects we study have variable incentive payments to task operators within the project company, and a general precedence structure.

This paper describes an IC mechanism to resolve Parkinson's Law and Student Syndrome for a project that is planned and executed by internal task operators under the critical path method. We show that our mechanism is group-strategy-proof. We also design an IC mechanism to resolve Parkinson's Law for a project that is planned under CCPM. We provide sufficient conditions on this mechanism such that the incentive payments received by all task operators are typically at least as much as those under the critical path method; moreover, the minimum guaranteed payment to

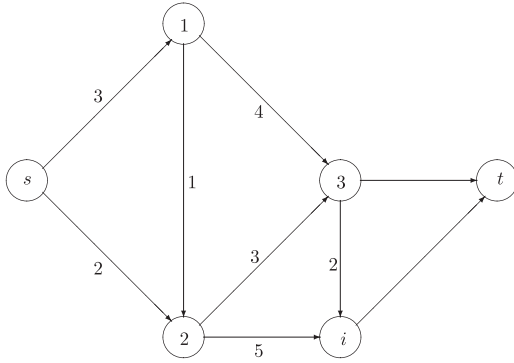


Fig. 1. A milestone and its contributing tasks.

the project manager remains unchanged. We develop an IC mechanism for multiple repeated projects planned under CCPM. Computational testing of our IC mechanisms reveals that both makespan and incentive payments are significantly reduced relative to the benchmark scheme most commonly used in practice, and our incentive mechanism performs effectively for multiple projects. All proofs appear in an Appendix.

2. Preliminaries

Section 2.1 describes the project network. Section 2.2 describes the information flow in the network. Section 2.3 describes the process by which the tasks are executed. Section 2.4 describes contractual and reporting requirements on the PM and TOs. Section 2.5 describes our model of project planning under CPM and CCPM.

2.1. Precedence network

A project controlled by a *project manager* (PM) consists of a number of tasks that are performed by *task operators* (TOs) who are employees of the project company. We initially assume that each TO operates one task, but we later relax this assumption. For distinction, we refer to the PM as “she” and to a TO as “he”. The overall project network, including the precedence relationships among the tasks, is modeled as a directed acyclic graph $G = (N, A)$, where the node set N represents *events* and the arc set A represents the tasks. We assume G is public knowledge. Denote $n = |A|$.

We particularly consider a subset $M \subseteq N$ of events that are designated as *milestones* of the project (Abu-Hilje & Ibbs, 1989). Milestones represent critical events within the progress of the project, for example, the completion of a project module. We define milestones as events where the PM receives revenue, i.e., a partial payment from the project client. From the project contract, this revenue may depend on the completion time of the milestone. For any project, $M \neq \emptyset$, since there is always a milestone representing the completion of the whole project. For any milestone $i \in M$, we let $G_i = (N_i, P_i)$, for $i = 1, \dots, m = |M|$, denote the subgraph of G induced by the nodes $N_i \subseteq N$, including i , from which there is a directed path to node i . Each task $j \in P_i$ is a *contributing task* for milestone i , for $i \in M$. For each task $j \in A$, we denote by Q_j the set of all predecessors of task j .

Example 1. The graph in Fig. 1 represents a project with $n = 9$ tasks. There are two milestones, i and t , where t indicates the completion of the project, thus $M = \{i, t\}$. Milestone i has seven contributing tasks, represented by arcs with task durations shown. □

2.2. Private information about ready time and duration

We model uncertainty in the task durations as follows. We assume that, for each task $j \in A$, its duration Δ_j satisfies an *unknown* discrete probability distribution with support $S_j(0) = \{a_j, a_j + 1, \dots, b_j\}$, which is public knowledge. Let $\bar{T}$ be the project completion time calculated from task duration upper bounds $\{b_j: j \in A\}$. At time $t \leq \bar{T}$ after task j is started, a more accurate estimate for the support of Δ_j , denoted by $S_j(t) = \{a_j(t), a_j(t) + 1, \dots, b_j(t)\} \subseteq S_j(0)$, may be obtained as information private to TO j . We assume support $S_j(t)$ is non-increasing in t in terms of set containment, reflecting that the process of executing the task can improve but not worsen the accuracy of the estimate for the task duration Δ_j . On the other hand, for each task $j \in A$, there is a time-dependent *ready time* $r_j(t)$ at which the TO is ready to execute task j , unless any of his predecessor tasks in Q_j are incomplete. Note that $r_j = r_j(0)$ is computed from task duration upper bounds $\{b_j: j \in A\}$. Since the TOs work for the project company, the values of a_j , b_j and r_j are set by the PM at the planning stage. Margraf (2018) summarizes six sources of information for doing so: work breakdown, historical data, analogous tasks, expert judgement, effort involved, and units of activity. However, for any $t = 1, \dots, \bar{T}$, knowledge of $r_j(t)$ before task j starts is private to TO j only.

It is possible that, at some integer time point $t > 0$ before task j starts, TO j may become ready earlier than r_j (i.e., $r_j(t) < r_j$) to execute his task j . In this case, he can choose to report to the PM an *earlier* integer ready time $\bar{r}_j(t)$ with $r_j(t) \leq \bar{r}_j(t) < r_j$. Similarly, at some time point $t > 0$ after task j has actually started, the TO may have less uncertainty about the duration of his task and he can choose to report to the PM a *reduced* maximum integer duration $\bar{b}_j(t)$ with $b_j(t) \leq \bar{b}_j(t) < b_j$. We provide incentives for TO j to make such reports, with the requirement that the reported $\bar{r}_j(t)$ and $\bar{b}_j(t)$ values are binding. Thus, the agreed r_j and b_j values are updated by $\bar{r}_j(t)$ and $\bar{b}_j(t)$, respectively, from time t onwards.

For any $j \in A$, assume an updated report $\bar{r}_j(t)$ (respectively, $\bar{b}_j(t)$) is made. We say that report $\bar{r}_j(t)$ (resp., $\bar{b}_j(t)$) is *truthful* if, on the one hand, $r_j(t) \leq \bar{r}_j(t) < \bar{r}_j(\bar{t})$ (resp., $b_j(t) \leq \bar{b}_j(t) < \bar{b}_j(\bar{t})$), where $\bar{t}$ ($0 \leq \bar{t} < t$) is the last time TO j reports a binding ready time $\bar{r}_j(\bar{t})$ (resp., duration $\bar{b}_j(\bar{t})$) or no such report has been made (i.e., $\bar{t} = 0$ with $\bar{r}_j(0) = r_j$ and $\bar{b}_j(0) = b_j$); and on the other hand, the report is the *smallest* estimate *without* the risk of future violation of the binding requirement.

Starting with the initial setting of $\bar{r}_j(t) = r_j$, $\bar{a}_j(t) = a_j$, $\bar{b}_j(t) = b_j$ for all $j \in A$ and all $t \in \{0, 1, \dots, \bar{T}\}$, the project proceeds for $t = 1, \dots, \bar{T}$ with possible reported updates $\{\bar{r}_j(t), \bar{b}_j(t)\}$. Then both $\bar{r}_j(t)$ and $\bar{b}_j(t)$ are nonincreasing, which we will further explain in detail in Section 2.4.

2.3. Project execution

The *accessible time* $\xi_j(t)$ of task j at time t is the updated event time at which all predecessors Q_j of task j will have been completed, based on the latest reported information, i.e.,

$$\xi_j(t) = \max_{k \in Q_j} \{\bar{r}_k(t) + \bar{b}_k(t)\}.$$

The accessible time of task j is the time at which task j can start, provided TO j is ready, and is clearly nonincreasing in t since both $\bar{r}_k(t)$ and $\bar{b}_k(t)$ are nonincreasing for all $k \in Q_j$. This definition identifies an information asymmetry, where the PM knows the accessible time of task j , but does not know if task j is ready until TO j reports so.

Recall that, for each event $i \in N$, we use $P_i \subseteq A$ to denote all tasks contributing to event i . Denote by $\bar{c}_i = \max_{j \in P_i} \{r_j + b_j\}$ the earliest time in the planned schedule, using task duration upper bounds as

estimates, by which all tasks in P_i will have been completed, and let

$$C_i = \max_{j \in P_i} \{\bar{r}_j(\bar{C}_i) + \bar{b}_j(\bar{C}_i)\}. \quad (1)$$

Eq. (1) defines C_i as the time at which milestone i is achieved during project execution.

During the execution of task j , TO j may improve his estimated task duration from b_j to $b_j(t) < b_j$. Further improvements may occur later. However, as a result of Parkinson's Law, TO j may decide not to report an earlier ready time or a shorter task duration, and this typically worsens project performance. Another situation, also within the scope of Parkinson's Law but behaviorally different, is that a TO works more slowly than necessary. However, from the perspective of the PM and the other TOs, these situations are mathematically identical, since successor tasks cannot start until the current task is reported as completed.

2.4. Contractual and reporting requirements

In addition to the precedence constraints in $G = (N, A)$, we impose the following requirements.

1. *Contractual constraints:* The initial task ready time $r_j = r_j(0)$ and task duration $b_j = b_j(0)$ are contractually binding on the TOs. Since the TOs are employees of the company, these upper bounds are agreed at the planning stage.
2. *Certification of completion:* A TO cannot fake the completion of his task, which is verified by an independent expert. This is common in, for example, environmental projects (Alberini, Longo, Tonin, Trombetta, & Turvani, 2005). Further, in any project, certification of completion is important for internal quality control, accurate allocation of costs and performance incentives, accurate reporting of progress to the client, and initiation of milestone payment requests.

Initially, the PM has only the original information $r_j(0)$ and $b_j(0)$. Before the whole project is completed and at each time $t = 1, \dots, \bar{T}$, we define task set $A(t) = \{j \in A : \bar{r}_j(t-1) > t\}$ and let the following information exchange take place:

1. *Completion reports.* Each TO j is required by contract to report the completion of task j no later than time point $\bar{r}_j(t-1) + \bar{b}_j(t-1)$. This requirement is common in environmental projects (Alberini et al., 2005).
2. The PM announces updated accessible times $\xi_j(t-1)$ to all those TOs j who have not made their Completion Reports.
3. Each TO j , who has not made a Completion Report, has the opportunity to make an update report of $\bar{r}_j(t)$ (if $j \in A(t)$) and $\bar{b}_j(t)$ based on his private information of $r_j(t)$ and $b_j(t)$. If he does so, the new ready time and duration bound are binding and satisfy the following requirements: $\bar{b}_j(t) \leq \bar{b}_j(t-1)$ and, if $j \in A(t)$,

$$\xi_j(t-1) \leq \bar{r}_j(t) \leq \bar{r}(t-1). \quad (2)$$

By reporting a new ready time $\bar{r}_j$, TO j accepts the requirements (a) to start task j no later than time $\bar{r}_j(t)$, and (b) to complete task j no later than $\bar{r}_j(t) + \bar{b}_j(t)$, by the Contractual Constraint. If no update is reported, then the existing values are used: $\bar{r}_j(t) = \bar{r}_j(t-1)$ and/or $\bar{b}_j(t) = \bar{b}_j(t-1)$.

2.5. Modeling CPM and CCPM

We define project planning under CPM, consistently with Wysocki (2009) and Klatorin (2012), as follows. Historical data is collected from all the task operators within the project company. Based on this data, estimates are formed of an optimistic duration and a pessimistic duration for each task. Even for tasks with limited data, the project management literature (Golenko-Ginzburg,

1988; Kerzner, 2009) recognizes the ability of a project manager to do this.

In a CPM system, TOs often overestimate the time that their tasks will take (Goldratt, 1997), in order to avoid the appearance of mismanagement and avoid blame if the task duration is longer than expected. A study by Hill, Thomas, and Allen (2000) finds that, for small software development tasks, the estimated durations are more than 40% greater than actual durations. Bukhari and Malik (2012) find that overestimation is more frequent than underestimation. Thus, we model CPM using the 100th percentile, i.e., the smallest duration to which the TO can commit with complete confidence. Hence, using the pessimistic task durations b_j , $j = 1, \dots, n$, the PM applies CPM to compute a deadline for each task. The project starts at time 0 with a deadline of $\bar{T}$, as defined in Section 2.2. Also, the ready time of each task is computed as the latest completion time of any of its predecessor tasks. Based on information collected from all the TOs within the organization, the PM informs each TO about the initial ready time and deadline of his task.

We define project planning under CCPM, consistently with Goldratt (1997), as follows. Similar to CPM, estimates of an optimistic duration a_j and a pessimistic duration b_j are formed for each task. In a typical implementation of CCPM, a mid-range estimate of task duration, such as the median estimate, is used to limit unnecessary slack in the project (Herroelen & Leus, 2001; Leach, 1999). Lacking a probability distribution for task duration, we use $m_j = (a_j + b_j)/2$ as a proxy measure for the median. The PM makes TO j aware that the project is planned using the m_j value. Recall that, under CCPM, task deadlines are not defined. Leach (1999) notes "Critical chain project plans only provide dates for the start of activity chains and the end of the project buffer".

2.6. Formal problem definition

We formulate our problem as one of mechanism design. We omit notational dependencies where they are clear from context. Let $\Pi = (G, \mathbf{S}, \mathbf{Z})$ be a given project, where $G = (N, A)$ is the set of precedence constraints between tasks $j \in A$ represented by task owners (TOs), $\mathbf{S} = (S_j(0) : j \in A)$ provides the support $S_j(0) = \{a_j, a_j + 1, \dots, b_j\}$ for an uncertain duration Δ_j of each task $j \in A$, and $\mathbf{Z} = (Z_i : i \in M)$ specifies contractual payment Z_i to the PM for achieving each milestone $i \in M \subseteq N$. We wish to design a mechanism $\mathcal{M} = \mathcal{M}(\Theta)$, where Θ is a set of fixed parameters in a specified subspace $X \subset \mathbb{R}_+^k$ and k is some positive integer, for the PM to receive from every TO $j \in A$ an updated report of his task ready time $\bar{r}_j$ and task duration $\bar{b}_j$. The objective is to minimize the makespan $\zeta = \zeta(\mathbf{Z})$ of the project Π , where the TOs' reports $\mathbf{z} = ((\bar{r}_j, \bar{b}_j) : j \in A)$ are based on their private information $\mathbf{z}(t) = ((\bar{r}_j(t), \bar{b}_j(t)) : j \in A)$ as described in Section 2.2, and are subject to the constraints described in Section 2.4. The mechanism $\mathcal{M}(\Theta)$ specifies the amount $p_{ij}(\mathbf{z}, \Theta)$ of payment by the PM to every TO $j \in A$, subject to the constraint that $\sum_{j \in A} p_{ij}(\mathbf{z}, \Theta) \leq Z_i$ for each milestone $i \in M$. As discussed in Section 2, this problem reduces to designing a mechanism $\mathcal{M}(\Theta)$ that is truthful. That is, under such a mechanism, the TOs each maximize their total milestone payments by reporting the true values, as defined in Section 2.2, of their ready times and task durations.

Let $\zeta_{\min}(\Theta)$ be the minimum makespan of project Π under mechanism $\mathcal{M} = \mathcal{M}(\Theta)$. It is clear that the value $\zeta_{\min}(\Theta)$ is a constant $\zeta_{\min}$, independent of Θ , as long as $\mathcal{M}(\Theta)$ is truthful, which we establish below for all possible $\Theta \in X$. As a further step, we explicitly choose the values of parameters $\Theta^* \in X$, so that the total of the PM's retained milestone payment, $\sum_{i \in M} (Z_i - \sum_{j \in A} p_{ij}(\mathbf{z}, \Theta))$, is maximized at Θ^* . Thus, under truthful mechanism $\mathcal{M}(\Theta^*)$, the PM minimizes the project makespan to $\zeta_{\min}$ and, subject to

achieving this objective, she retains as much of the milestone payments as possible.

3. Mechanisms for CPM

This section describes our two mechanisms for a project that is planned using CPM. Section 3.1 describes our mechanism, and establishes that it is incentive compatible. Section 3.2 designs a more general “weighted” scheme to incentivize TOs who own multiple dependent tasks. Section 3.3 shows how optimal parameters for both our incentive schemes can be found. The weighted incentive scheme is evaluated computationally in Section 3.4 against a benchmark scheme.

3.1. Incentives

Assume for this subsection that a TO owns one task. We consider a milestone payment from the project client, denoted by Z_i for any fixed $i \in M$, and how to share it among the PM and the TOs. We let

$$Z_i = \rho_i \cdot (\bar{C}_i - C_i), \quad (3)$$

where $\rho_i > 0$ is a constant. Observe that $(\bar{C}_i - C_i) > 0$ represents the amount of improvement in milestone completion time. Then, ρ_i converts this improvement into a contractual bonus paid by the project owner, and can be viewed as a *per-unit rate* of payment for performance at milestone i . However, any monotonically increasing function $f(\bar{C}_i - C_i)$ which ensures that earlier project completion is better suffices. A task may contribute to several milestones, in which case its shares of milestone payments are calculated separately and sum to its total payment. We describe our incentive scheme that allows flexibility based on chosen parameters λ_{ij} and μ_{ij} . Here, $0 < \lambda_{ij} < 1$ converts both a saving in task duration and an earlier ready time into a positive amount of slack, as defined below, for task j of milestone i . Also, $\mu_{ij} > 1$ specifies the relative importance to slack of savings in ready time and savings in duration and ensures that there is no trade-off between the two. For each task $j \in A$, our incentive scheme incentivizes earlier submission of reports of achievable reduced ready time $\bar{r}_j(t)$ and task duration $\bar{b}_j(t)$.

For each $j \in A$, let $T_j \in \{1, \dots, \bar{T}\}$ be the reported completion time of task j , where $T_j \leq r_j + b_j$, and denote by

$$r'_j = \bar{r}_j(T_j) \quad \text{and} \quad b'_j = \bar{b}_j(T_j)$$

the last reported, if any, ready time and duration of task j , respectively. Let τ_j^r and τ_j^b denote the corresponding times, if any, when r'_j and b'_j , respectively, are reported. Thus,

$$\tau_j^r = \begin{cases} \min\{t \leq T_j : \bar{r}_j(t) = r'_j\}, & \text{if } r'_j < r_j; \\ r_j, & \text{if } r'_j = r_j; \end{cases}$$

$$\tau_j^b = \begin{cases} \min\{t \leq T_j : \bar{b}_j(t) = b'_j\}, & \text{if } b'_j < b_j; \\ T_j, & \text{if } b'_j = b_j. \end{cases}$$

We use $f_j = (\tau_j^r + \tau_j^b)/(2r_j + b_j)$ as a metric for early reporting, where $0 < f_j \leq 1$.

(λ, μ) -Scheme

1. For any milestone $i \in M$ with $Z_i > 0$, fix parameters λ_{ij} and μ_{ij} with $0 < \lambda_{ij} < 1 < \mu_{ij}$. Compute the positive slack of any task $j \in P_i$ as

$$s_j = \mu_{ij}(r_j - \lambda_{ij}r'_j) + (b_j - \lambda_{ij}b'_j). \quad (4)$$

2. Denote $D_i = \sum_{k \in P_i} (\mu_{ik}r_k + b_k)$. Make an incentive payment on task $j \in P_i$ of

$$p_{ij} = \frac{s_j + D_i\omega(1 - f_j)}{D_i(1 + \omega)} (\sigma_i Z_i), \quad (5)$$

where $\omega \geq 0$ is a chosen parameter that controls the importance of an early report relative to an early start or early completion. For example, $\omega = 0$ indicates that it is negligible, whereas $\omega \rightarrow +\infty$ indicates that it is dominant. Also, $\sigma_i \in (0, 1]$ is another chosen parameter which we discuss in detail in Section 3.3.

3. The PM keeps the remainder of the payment Z_i for milestone i :

$$\frac{\sum_{k \in P_i} (\lambda_{ik}(\mu_{ik}r'_k + b'_k) + D_i\omega f_k)}{D_i(1 + \omega)} (\sigma_i Z_i). \quad (6)$$

End of scheme

Remark 1. For any fixed milestone $i \in M$ and fixed task $j \in P_i$, parameter λ_{ij} controls the sensitivity to slack s_j , and hence to the amount of incentive based on it, of an earlier ready time r'_j and a reduced task duration b'_j . Also, parameter μ_{ij} controls the relative importance of an earlier ready time to a reduced task duration.

A numerical example of the (λ, μ) -Scheme appears in Section A.1 of the Appendix.

Remark 2. In the (λ, μ) -Scheme, the PM informs TO $j \in P_i$ of the values of λ_{ij} and μ_{ij} before project execution.

The next theorem establishes bounds on the payments received by the various parties.

Theorem 1. For each milestone M_i for which $Z_i > 0$, with the incentive payments defined in the (λ, μ) -Scheme, the PM receives a fraction of the milestone payment $\sigma_i Z_i$ between $\kappa_i \lambda_i^{\min}/(1 + \omega)$ and $(\lambda_i^{\max} + \omega)/(1 + \omega)$, where

$$0 < \kappa_i = \frac{\sum_{k \in P_i} a_k}{\sum_{k \in P_i} (\mu_{ik}r_k + b_k)} < 1,$$

and $\lambda_i^{\min} = \min_{j \in P_i} \lambda_{ij}$ and $\lambda_i^{\max} = \max_{j \in P_i} \lambda_{ij}$, while the TOs collectively receive a fraction of the total payment $\sigma_i Z_i$ between $1 - (\lambda_i^{\max} + \omega)/(1 + \omega)$ and $1 - \kappa_i \lambda_i^{\min}/(1 + \omega)$. Further, the (λ, μ) -Scheme is budget balanced.

Theorem 1 establishes how the choice of incentive parameters modulates the shares of the available incentive received by the various parties. These shares can be adjusted by the PM, based on the employment incentives of the TOs. We now provide the main result of this section.

Theorem 2. The (λ, μ) -Scheme is incentive compatible for TO j to report his “true values” (as defined in Section 2.2) of r'_j and b'_j as early as possible, for any milestone $i \in M$, and $j \in P_i$.

Remark 3. The use of two parameters λ_{ij} and μ_{ij} , instead of one, separates the reports on r_j and b_j to avoid strategic trading off of the two by TO j . Further, these parameters can be used to adjust the relative payments received by each TO j and the PM, for any fixed milestone $i \in M$. Moreover, these values can be varied across different milestones $i \in M$.

Remark 4. In the (λ, μ) -Scheme, Steps 1, 2 and 3 each require $O(n)$ time, for each $i \in M$. Hence, the overall complexity of the scheme is $O(nm)$ time.

3.2. Extensions

In this section, we study how the result in Theorem 2 can be generalized to allow the PM to assign multiple tasks to the same TO, as occurs in many organizations. The next result provides such a generalization, under a specific condition.

Corollary 1. The (λ, μ) -Scheme is incentive compatible, even if the PM assigns multiple tasks to a TO, provided no such task is a predecessor of another.

From [Corollary 1](#), if several tasks without dependence among them are assigned to the same TO, there is no advantage to the TO to make decisions for them jointly. This gives the PM substantial flexibility in assigning tasks to owners. However, the following example shows that the (λ, μ) -Scheme fails when a TO owns several tasks and there is dependence among them.

Example 2. Consider events $\{s, 1, \dots, n, t\}$ and tasks $\{(s, 1), (1, 2), \dots, (1, n), (2, t), \dots, (n, t)\}$. All tasks have an execution time in $[a, 2a]$. Suppose that a TO owns all the tasks and realized durations are $2a$ for all tasks except $(s, 1)$ which requires time a . In this case, he receives an incentive payment of $\bar{\alpha} \cdot a$ as a result of being truthful, where $\bar{\alpha} > 0$ is the per diem bonus rate for early completion. However, by falsely reporting all task durations as $2a$ except for tasks $(1, t), \dots, (n, t)$ which are reported as a , he receives an incentive payment of $n \cdot \bar{\alpha} \cdot a$. This false reporting is possible since he can report the completion of task $(s, 1)$ as $2a$, but start the tasks of type (i, t) right after the true completion time, a , of $(s, 1)$. Hence, false reporting provides an increase in incentive payment to the TO. However, it may also increase the project makespan since other project tasks that are successors of task $(s, 1)$ cannot start until time $2a$ when the completion of that task is reported falsely late. $\square$

To resolve this problem, we describe a generalized incentive payment scheme, the *Weighted (λ, μ) -Scheme*. Intuitively, we provide larger incentive weights to the tasks with more successors, so that the possibility of increasing incentive payment due to dependence between the tasks is offset by the weight each task carries.

Weight assignment

For each task $j \in A$, let V_j denote the set of *dependent tasks* of j . That is, if $j = (j_{\text{tail}}, j_{\text{head}}) \in A$, then V_j is the set of tasks to the tail of which there is a directed path from the head j_{head} of j . We define a weight function as follows: if $V_j = \emptyset$, let $w_j = 1$; otherwise, let $w_j = 1 + \sum_{k \in V_j} w_k$. Observe that each task $j \in A$ is assigned a weight that is strictly greater than the sum of the weights for all its successors in V_j . Since the tasks form a directed acyclic graph, the weights are well defined.

Weighted (λ, μ) -Scheme

1. Compute weights $\{w_j : j \in N\}$, and inform each TO of the weight(s) of his task(s). For any milestone $i \in M$ with $Z_i > 0$, fix parameters λ_{ij} and μ_{ij} with $0 < \lambda_{ij} < 1 < \mu_{ij}$. Let $\tilde{D}_i = \sum_{k \in P_i} w_k (\mu_{ik} r_k + b_k)$. Make a payment to TO $j \in P_i$ as follows:

$$p_{ij} = \frac{w_j s_j + \tilde{D}_i \omega (1 - f_j)}{\tilde{D}_i (1 + \omega)} (\sigma_i Z_i), \quad (7)$$

where s_j , ω and σ_i are as in the (unweighted) (λ, μ) -Scheme.

2. The PM keeps the remaining positive fraction of $\sigma_i Z_i$ (see [Section 3.3](#)).

End of scheme

A numerical example of the Weighted (λ, μ) -Scheme, appears in [Section A.5](#) of the Appendix. The next result establishes that the Weighted (λ, μ) -Scheme is incentive compatible, even if a TO owns some tasks that are predecessors of others.

Theorem 3. Assume that we are given a directed acyclic graph $G = (N, A)$ of events and tasks, and an arbitrary assignment of tasks to TOs. Then, the Weighted (λ, μ) -Scheme is incentive compatible for any TO to report as soon as possible the “true values” (as defined in [Section 2.2](#)) r'_j and b'_j of his jobs j .

A mechanism is *strongly group-strategy-proof* if it induces a non-cooperative game in which no group of players can collude to misreport their private information in a way that makes at least one

member of the group better off without making any of the remaining members worse off. The following corollary about the strong group-strategy-proofness of the Weighted (λ, μ) -Scheme follows immediately from [Theorem 3](#).

Corollary 2. The Weighted (λ, μ) -Scheme is strongly group-strategy-proof.

To see the correctness of the above corollary, assume to the contrary that there is such a colluding group in the reporting game played by all TOs. Consider another reporting game in which all tasks of the members of the colluding group are assigned to a single TO, while assignment of the other tasks remains the same. Then in this new reporting game, the “new” TO can misreport his private information to make himself better off, which contradicts [Theorem 3](#).

The following result is evident.

Corollary 3. Since the TOs are incentivized both to start and finish their tasks as soon as possible under the (λ, μ) -Scheme and the Weighted (λ, μ) -Scheme, [Theorems 2](#) and [3](#) also resolve Student Syndrome. $\square$

In our work, we include both the weighted and unweighted versions of our (λ, μ) -Scheme. The unweighted version is easier to understand and implement. However, the weighted version enables the PM to deal with more general and flexible assignment of tasks to the TOs, while still retaining incentive compatibility. This additional flexibility can be used both to alleviate resource bottlenecks in the project, and to implement secondary criteria in rewarding the TOs.

3.3. Optimal choice of scheme parameters

From the perspective of mechanism design, the PM is the principal who designs the IC mechanisms. The PM derives disutility from a longer makespan, as a result of specifications in the contract with the project owner. Hence, her primary objective is to resolve the effect of Parkinson's Law on the project in order to minimize the makespan. As explained in [Section 2.6](#) above, the PM has a secondary objective of retaining as much of the milestone payments as possible, while providing them in sufficient amount to incentivize the TO employees to report truthfully. That is, among all the payment schemes that achieve the minimum project makespan, she has a preference for one that retains as much of the resulting milestone payments as possible. Such a payment scheme can be found by optimizing the payment scheme parameters.

3.3.1. Optimal parameters in the (λ, μ) -Scheme

For simplicity but without significant modification to the problem, in the (λ, μ) -Scheme, we set $\lambda_{ij} = 1/2$ for any milestone i and any TO j and set $\omega = 0$ (see the ranges and interpretations of these parameters in the description of the (λ, μ) -Scheme in [Section 3.1](#)). We assume $Z_i > 0$ and let $\mu_{ij} = \mu_i$ for any j , which indicates that all contributing tasks to milestone i are treated equally in terms of sensitivity to the incentive payment of an earlier ready time r'_j and a reduced task duration b'_j . Then, from (4), we find that a one time unit improvement in ready time r_j (i.e., from r_j to $r'_j = r_j - 1$) and in task duration b_j , respectively, lead to the following fractions of Z_i being paid to TO j :

$$p_{ij}^1 = \frac{\mu_i (r_j + 1) + b_j}{2\mu_i R_i + 2B_i} \sigma_i \quad \text{and} \quad p_{ij}^2 = \frac{\mu_i r_j + (b_j + 1)}{2\mu_i R_i + 2B_i} \sigma_i, \quad (8)$$

where $R_i = \sum_{k \in P_i} r_k$ and $B_i = \sum_{k \in P_i} b_k$. Note that $p_{ij}^1, p_{ij}^2 \rightarrow q_{ij} \sigma_i$ as $\mu_i \rightarrow 1$, which indicates that one unit of earlier ready time has the

same incentive payment as one unit of shorter task duration for all tasks $j \in P_i$, where

$$q_{ij} = \frac{r_j + b_j + 1}{2R_i + 2B_i}.$$

Let $Q_i = \sum_{j \in P_i} q_{ij}$, where Q_i can be viewed as the total fraction of compensation to the TOs who contribute to milestone i . Clearly, $1/2 < Q_i < 1$. Suppose that in (3) the overall rate ρ_i of payment on milestone i , which could be negotiated by the PM with the project client, is such that it is guaranteed to motivate all TOs $j \in P_i$ if their corresponding individual rates sum up to $\rho_i/2$, i.e., $Q_i \sigma_i = 1/2$. Now, the PM finds the *minimum* value of parameter $\sigma_i \in (0, 1]$ to provide sufficient motivation for any TO $j \in P_i$ to report truthfully, given that this is guaranteed by setting $\sigma_i = 1/(2Q_i) < 1$. In practice, these values can be set from industry standards, or from employment contracts and negotiations with the TOs.

In general, when desirable values of parameters λ , μ and ω are fixed and an adequate value of ρ_i has been negotiated, the PM minimizes parameter $\sigma_i \in (0, 1]$ subject to the condition that the payment defined in (5) provides sufficient incentive for every TO $j \in P_i$ to report truthfully.

3.3.2. Optimal parameters in the Weighted (λ, μ) -Scheme

With the same setting of ρ_i , λ_{ij} , μ_{ij} and ω as for the unweighted (λ, μ) -Scheme, we apply the same arguments for optimizing σ_i , taking into account the values $\tilde{Q}_i = \sum_{j \in P_i} \tilde{q}_{ij}$, where

$$\tilde{q}_{ij} = \frac{w_j(r_j + b_j + 1)}{2 \sum_{k \in P_i} w_k(r_k + b_k)}.$$

The optimization procedure described in the last paragraph of Section 3.3.1 also applies here.

3.4. Incentive scheme performance

We compare the performance of our Weighted (λ, μ) -Scheme with that of a widely used benchmark incentive scheme described in Section 3.4.1. Our computational study is reported in Section 3.4.2.

3.4.1. Benchmark

During the long history of project management practice, various incentive schemes have been developed to improve project performance. These schemes can be classified as improving project performance with respect to (a) schedule, (b) cost, (c) quality, (d) safety, or (e) combinations of these (Bubshait, 2003; Ibbs, 1991). For relevance to our discussion of Parkinson's Law, we focus on schedule incentive schemes.

Several studies of business practice highlight the importance of schedule incentives in improving project performance. Arditi and Yasamis (1998) study 21 Illinois Department of Transportation highway contracts, and conclude that the projects would have taken 21% longer on average to complete without schedule incentive schemes. Kog, Chua, Loh, and Jaselskis (1999) use a neural network approach to study the effectiveness of schedule incentive schemes in 20 projects. They conclude that an incentive of 3–5% of project value is effective at improving on-time performance, especially where other aspects of project support are poor. Radulescu and Heales (2005) survey 117 information systems project managers. Among the respondents, 63% believe that schedule incentive schemes improve project delivery times.

The key characteristics of schedule incentive schemes include (a) whether they are implemented through a contract price adjustment, a side payment, or a profit sharing scheme (Bower, Ashby, Gerald, & Smyk, 2002), (b) whether they are based on individual milestones or on overall completion time (Ibbs, 1991), and (c) whether the incentive is positive as with a bonus, or negative as

Table 1

Computational results for the Weighted (λ, μ) -Scheme with one falsely reporting TO.

n	OS	Imp	Imp-P	Imp-M	PY (%)	MK (%)
50	0.1	768	748	304	38.91	5.59
50	0.2	836	809	497	49.81	5.66
50	0.3	812	788	533	52.98	6.13
100	0.1	881	862	459	55.18	5.15
100	0.2	889	865	572	63.50	5.10
100	0.3	827	795	559	59.05	5.13
200	0.1	888	885	203	61.88	5.79
200	0.2	821	815	256	61.89	6.02
200	0.3	752	742	241	53.66	6.51
Mean		830	812	403	55.21	5.68

with a penalty (Herten & Peeters, 1986). For comparability, we consider profit sharing schemes based on individual milestones, and allow both positive and negative incentives. The related literature (Abu-Hilje & Ibbs, 1989; Arditi & Yasamis, 1998; Bayiz & Corbett, 2005; Bower et al., 2002) identifies only two such schedule incentive schemes that are widely used:

Scheme 1: A per diem bonus for early completion, or penalty for late completion;

Scheme 2: A lump sum bonus for on-time completion.

Scheme 1 is applicable where early completion provides additional value to the project owner (Abu-Hilje & Ibbs, 1989; Arditi & Yasamis, 1998), as in the problem we are studying. In the schemes we propose, there is no bonus for on-time completion. This is because completion of a task is guaranteed by the contractual requirement that the duration of each task must not exceed its upper bound, which is a conservative estimate of the task duration. Therefore, in our comparison of incentive scheme performance in Section 3.4.2, we consider only Scheme 1. As shown in Example 2, Scheme 1 is vulnerable to false reporting, which we formalize in Section A.8 of the Appendix. We now study the impact of this vulnerability on project performance.

3.4.2. Computational study

We conduct a computational study to compare the effectiveness of the Weighted (λ, μ) -Scheme and Scheme 1, at reducing the makespan and the total payments to the TOs. Since Scheme 1 does not reward earlier ready times and earlier reports, we also ignore ready times and remove any incentives for early reporting in the Weighted (λ, μ) -Scheme. Hence, we set $\mu = 0$ and $\omega = 0$. Recall that in our Weighted (λ, μ) -Scheme, we require that $\mu > 1$ and $\omega > 0$. This is because that scheme does not ignore ready times and does provide incentives for early reporting. For simplicity, we set $\rho_i = \sigma_i = 1$ in the Weighted (λ, μ) -Scheme. Then, to provide a fair comparison, in the Weighted (λ, μ) -Scheme we set $\lambda = 1/2$, and in Scheme 1 we set the per diem bonus rate at $\bar{\alpha} = 1/2$ and the per diem penalty rate at $\bar{\beta} = +\infty$. Then in both schemes the reported duration of task j does not exceed b_j , and the bonus rates for early completion are the same.

We first consider the simplest scenario of only one falsely reporting TO, with all other TOs having realized task durations at their upper bound values b_j . Then, we consider multiple falsely reporting TOs. For the case of single falsely reporting TO, we generate random data as follows: (a) The underlying task precedence networks are generated with RanGen (Demeulemeester, Vanhoucke, & Herroelen, 2003); (b) for all tasks j , $a_j \sim \text{UI}[1, \dots, n/2 + 1]$, $b_j \sim \text{UI}[a_j + 1, \dots, a_j + n/2 + 1]$; and (c) for one single task k of the falsely reporting TO in each instance, its realized duration is $b'_k \sim \text{UI}[a_k, \dots, b_k]$. Our computational results are summarized in Table 1, where for each row except the last one we generate 1000 random instances. The final row shows mean results, computed

over the 9000 instances in the study. The seven columns are explained below, where Columns 4 and 5 provide more details about Column 3, and the last two columns are performance indicators averaged over the 1000 instances.

- “n” indicates the number of tasks in an instance.
- “OS”, which stands for *Order Strength*, is a measure of the density of the task precedence network (Demeulemeester et al., 2003).
- “Imp” shows the number of instances out of 1000 in which the Weighted (λ, μ) -Scheme improves on Scheme 1 with respect to at least one of the two criteria mentioned above, i.e. reduction of makespan or of total incentive payments to the TOs.
- “Imp-P” shows the number of projects out of 1000 instances that have reduction of total incentive payment under the Weighted (λ, μ) -Scheme, relative to Scheme 1.
- “Imp-M” shows the number of projects out of 1000 instances that have reduced project makespans under the Weighted (λ, μ) -Scheme, relative to Scheme 1.
- “PY” shows the percentage reduction in the total incentive payment under the Weighted (λ, μ) -Scheme relative to Scheme 1, averaged over all project instances where the total payments under the two schemes differ.
- “MK” shows the percentage reduction in the makespan under the Weighted (λ, μ) -Scheme relative to Scheme 1, averaged over all project instances where the makespans under the two schemes differ (i.e., all Imp-M instances). Observe that the makespan under the Weighted (λ, μ) -Scheme cannot exceed that under Scheme 1.

The results shown in Table 1 provide the following conclusions. First, even with only one task j being realized at less than its worst case estimate b_j , the Weighted (λ, μ) -Scheme delivers a reduction in either makespan or total payments to the TOs, or both, relative to Scheme 1, in about 83% of instances. Second, the total incentive payment to the TOs is typically about 55% less under the Weighted (λ, μ) -Scheme than under Scheme 1 when payments under the two schemes differ. Third, the Weighted (λ, μ) -Scheme typically captures nearly 6% more of the available reduction in makespan than Scheme 1, when one improved task duration reduces the makespan. Finally, all these results are robust across projects with various realistic sizes and network densities. These results collectively show that our Weighted (λ, μ) -Scheme is more effective at reducing project makespan and total incentive payments than the benchmark Scheme 1.

We also consider projects where multiple tasks, instead of only one, have realized durations that are less than their worst-case estimates. A corresponding computational study, which appears in Section A.9 in the Appendix, leads to similar conclusions. One overall conclusion from both sets of these results is that the Weighted (λ, μ) -Scheme resolves the problem of Parkinson's Law, which is in practice left unresolved by the widely used benchmark Scheme 1. A second overall conclusion is that the resulting savings in time and cost should be of significant benefit in many projects with widely varying characteristics.

4. Mechanism for CCPM

Advocates of CCPM (e.g., Steyn, 2001) claim that it alleviates Parkinson's Law. Nevertheless, task durations are still subject to estimates based on historical mid-range values (Herroelen & Leus, 2001). Hence, some of the inefficiencies attributable to Parkinson's Law, as discussed in Section 1, also apply under CCPM. To study this issue, in Section 4.1, we describe a mechanism for CCPM. Theorem 5 shows that this scheme is incentive compatible. In Section 4.2, Theorem 6 establishes sufficient conditions which ensure that the payments to the PM and also to each TO are the

same under CCPM as under CPM. Then, we define alternative conditions under which Theorem 7 compares the payments under the two planning systems.

4.1. Incentive scheme

We propose the following incentive scheme for projects planned under CCPM. For simplicity, we assume in this section that (a) all tasks are ready to start as soon as needed, (b) there is no bonus for earlier reports of reduced task durations, and (c) each TO owns one task or multiple but independent tasks. Our scheme uses parameters α , β and γ . Here, α_j (respectively, β_j) is a parameter that converts a gain in the duration of task j into a payment adjustment in the case where that duration is larger (resp., smaller) than the median duration m_j , and γ is a parameter that adjusts the value of an initial payment estimate to task j .

(α, β, γ) -Scheme

0. Fix any milestone $i \in M$. Let

$$\kappa^0 = \frac{\sum_{k \in P_i} a_k}{\sum_{k \in P_i} b_k}. \quad (9)$$

Fix γ such that $0 < \gamma \leq \kappa^0 < 1$ and, for each task $j \in P_i$, fix α_j and β_j such that

$$0 < \alpha_j < 1 < \beta_j, \quad (10)$$

and

$$\beta_j(1 - \gamma) < 1. \quad (11)$$

Define $m_j = (a_j + b_j)/2$. Let r'_j and b'_j denote the last reported ready times and durations.

1. Let $\tilde{Z}_i$ be the milestone payment based on the assumption that task j has a duration of m_j for each $j \in P_i$. Define

$$e_j = \frac{b_j - \tilde{\lambda} m_j}{\sum_{k \in P_i} b_k} \sigma_i \quad (12)$$

as the pre-execution estimated fraction of $\tilde{Z}_i$ as an incentive payment to TO j , where $0 < \tilde{\lambda} = 2\gamma/(1 + \kappa^0) < 1$ and $\sigma_i \in (0, 1]$ is a control parameter as in the (λ, μ) -Scheme.

2. If $b'_j \in [m_j, b_j]$, then make a payment to task j of $\phi_j e_j$ fraction of $\tilde{Z}_i$, where $\phi_j \in [\alpha_j, 1]$ is a monotonically decreasing function of b'_j :

$$\phi_j = \frac{(b_j - \alpha_j m_j) - (1 - \alpha_j) b'_j}{b_j - m_j}. \quad (13)$$

If $b'_j \in [a_j, m_j]$, then make a payment to task j of $\psi_j e_j$ fraction of $\tilde{Z}_i$, where $\psi_j \in [1, \beta_j]$ is a monotonically decreasing function of b'_j :

$$\psi_j = \frac{(\beta_j m_j - a_j) - (\beta_j - 1) b'_j}{m_j - a_j}. \quad (14)$$

3. The PM keeps the remaining fraction of $\tilde{Z}_i$:

$$1 - \sum_{j \in P_i} x_j e_j, \quad (15)$$

where $x_j \in \{\phi_j, \psi_j\}$ and hence $\alpha_j \leq x_j \leq \beta_j$.

End of scheme

Remark 5. Our assumptions (a), (b) and (c) for the (α, β, γ) -Scheme merely simplify the scheme. To remove restriction (a), we first calculate the pre-execution estimate of ready times $\tilde{r}_j$ using the midpoint durations m_j . Then, we define an “additional bonus” for reporting a ready time smaller than $\tilde{r}_j$ and a “loss of bonus” for reporting a ready time larger than $\tilde{r}_j$. To remove restriction (b),

we add into the bonus payment a suitably weighted term that uses the metric f_j for early reporting. To remove restriction (c), we introduce task weights as in the Weighted (λ, μ) -Scheme to prevent a TO from trading off incentive payments on earlier ready time and shorter task duration.

Our next result establishes bounds on the payments received by the various parties.

Theorem 4. *The (α, β, γ) -Scheme is budget balanced, and results in payments such that the TO of task j receives a fraction between $\alpha_j e_j$ and $\beta_j e_j$ of Z_i and the PM receives a fraction between $1 - \beta_{\max}(1 - \gamma) > 0$ and $1 - \alpha_{\min}(1 - \gamma) < 1$ of Z_i , where*

$$\alpha_{\min} = \min_{j \in P_i} \{\alpha_j\}, \quad \beta_{\max} = \max_{j \in P_i} \{\beta_j\}.$$

Remark 6. As noted in Remark 3, we can add a subscript i to all relevant parameters in the statement of the (α, β, γ) -Scheme and that of Theorem 4, such as to $\alpha_j, \beta_j, m_j, \phi_j, \psi_j$. These generalized parameters then adjust the relative payments received by the TOs and the PM for any fixed milestone $i \in M$, and also among different milestones in M . In practice, for example, a larger payment can be used to reward a TO with a longer tenure of employment.

The main result of this section is as follows.

Theorem 5. *The (α, β, γ) -Scheme for CCPM is incentive compatible.*

4.2. Comparing CPM and CCPM

We develop conditions to compare incentive payments under CPM planning with those under CCPM planning. We assume as in Section 4.1 that all tasks are ready to start as soon as needed, i.e., equalities are required to be satisfied in inequalities (2). Therefore, in our (λ, μ) -Scheme for the CPM system, there is no incentive payment for earlier ready time than r_j , and hence $\mu = 0$. Note that the usual requirement of $\mu > 1$ is used only when the ready times are considered for incentive payments. Therefore, $\kappa = \kappa^0$, as defined in (9). Now, with $0 < \lambda < 1$, we let

$$\gamma = \frac{1 + \kappa}{2} \lambda \leq \kappa, \quad (16)$$

and, for each $j \in P_i$, we let

$$\alpha_j = \frac{(1 - \lambda)b_j}{b_j - \lambda m_j}, \quad \text{and} \quad \beta_j = \frac{b_j - \lambda a_j}{b_j - \lambda m_j}. \quad (17)$$

Theorem 6. *Assume that all tasks are ready to start as soon as needed. In the (λ, μ) -Scheme for the CPM system (with $\mu = 0$) and the (α, β, γ) -Scheme for the CCPM system, with the publicly known parameters $\lambda, \{\alpha_j, \beta_j; j \in P_i\}$ and γ set using (16) and (17), then: (a) conditions (10) and (11) are satisfied, and (b) the same task duration reports $\{b'_j, j \in P_i\}$ lead to the same payments to each TO and the PM in the two incentive schemes, for any realized task durations.*

Let the parameters $\alpha_j, \beta_j, \gamma, \lambda$ and μ be set as in Theorem 6. We say that an (α, β, γ) -Scheme for the CCPM system is *enhanced* relative to a (λ, μ) -Scheme for the CPM system, if α_j and β_j in the (α, β, γ) -Scheme are replaced respectively by α and β for any $j \in P_i$, where

$$\alpha = \max_{j \in P_i} \{\alpha_j\}, \quad \text{and} \quad \beta = \max_{j \in P_i} \{\beta_j\}. \quad (18)$$

The next result then follows immediately.

Theorem 7. *Suppose an (α, β, γ) -Scheme is enhanced relative to a (λ, μ) -Scheme. Then with the same performance and reporting, each*

TO receives at least the same payment under the former scheme as under the latter scheme. On the other hand, the PM is guaranteed to retain a fraction $1 - \beta(1 - \gamma)\sigma_i$ of the total milestone payment Z_i .

Theorem 7, informs the PM about the choice of whether to use CPM or CCPM planning for a particular project. One interpretation of the theorem is that the TOs may receive a greater reward under CCPM. As a result, there is a preference for the PM to use CPM rather than CCPM for planning the project. However, this conclusion comes with two caveats. The first caveat is that the PM does not necessarily want to minimize the rewards gained by the employee TOs, for reasons of organizational morale and staff turnover. The second caveat is that the PM can always specify and achieve a guaranteed share of a milestone payment for herself under CCPM simply by adjusting the values of relevant parameters. Nonetheless, we believe that Theorem 7 provides the first analytical comparison between CPM and CCPM systems.

4.3. Optimal scheme parameters and computational study

Using Theorems 6 and 7, the PM can optimally choose the control parameters σ_i and other parameters in the enhanced (α, β, γ) -Scheme, as in the corresponding (λ, μ) -Scheme with $\mu = 0$ detailed in Section 3.3.

We now conduct a computational study of the incentive payments to the TOs specified by the (λ, μ) - and (α, β, γ) -Schemes for the CPM and CCPM planning systems, respectively. We are not aware of any previous literature that conducts such a comparison. Our study is motivated as follows. Recall from Theorems 2 and 5 that the (λ, μ) -Scheme and the (α, β, γ) -Scheme are incentive compatible under the CPM and CCPM planning systems, respectively. Hence, given identical realizations of the task durations, the project makespan values delivered by the two systems are identical. Therefore, we compare the total payments made to the TOs under the two systems in reaching the minimum achievable makespan.

As above, for simplicity we set $\sigma_i = 1$. Our data consists of nine combinations of parameters, specified by $n \in \{25, 50, 100\}$ and OS value $\in \{0.1, 0.2, 0.3\}$. These OS values are representative of many real project networks (Vanhoucke, Coelho, & Batse-lier, 2016). For each parameter combination, we randomly generate 20 instances, for a total of $9 \times 20 = 180$ instances, using RanGen (Demeulemeester et al., 2003). Consistent with our study in Section 3.4, in the simulation model only one task j has a realized duration that is potentially smaller than its upper bound b_j .

Table 2 reports results for the mean, median and standard deviation of the ratio, q , of payments to the TOs under the CCPM system for each instance, relative to those under the CPM system. First, consistent with Theorem 7, these ratios are at least 1 for every instance. The mean payment is on average 23.8% higher, and the median payment is on average 27.1% higher, under the CCPM scheme. The standard deviation of the payments is similarly 21.8% higher under the CCPM scheme. The ratio of mean and median payments increases slightly with n , does not vary significantly with OS, and increases strongly with λ .

These results provide the following useful insights to managers. First, the CCPM scheme provides additional flexibility in making payments to TOs. This flexibility can be useful in motivating particular individuals within the project company. Second, on average, the payments to the TOs are significantly higher under the CCPM system. However, the PM can enforce her guaranteed share of project revenue by setting λ appropriately, hence the additional payments to the TOs need not be problematic. Third, the variability of the payments to the TOs is typically greater under the CCPM scheme, however this difference is only proportional to the average

Table 2
Ratio ϱ of payments to TOs under CPM and CCPM.

n	OS	λ	ϱ mean	ϱ median	ϱ std. dev.
25	0.10	0.25	5.7	5.6	5.4
25	0.10	0.50	17.0	17.8	21.4
25	0.10	0.75	44.6	46.9	40.3
25	0.20	0.25	5.9	5.5	7.8
25	0.20	0.50	17.6	18.7	17.2
25	0.20	0.75	42.7	48.0	38.6
25	0.30	0.25	5.9	5.9	7.3
25	0.30	0.50	16.5	18.0	16.9
25	0.30	0.75	39.8	39.2	39.9
50	0.10	0.25	6.3	6.5	5.2
50	0.10	0.50	18.6	20.0	17.6
50	0.10	0.75	48.5	56.6	39.1
50	0.20	0.25	6.2	6.4	6.2
50	0.20	0.50	18.3	21.4	15.1
50	0.20	0.75	48.4	56.5	41.5
50	0.30	0.25	6.3	6.7	6.5
50	0.30	0.50	18.3	21.7	14.8
50	0.30	0.75	46.8	54.6	42.4
100	0.10	0.25	6.6	7.0	6.3
100	0.10	0.50	19.0	22.1	18.7
100	0.10	0.75	51.6	69.2	42.0
100	0.20	0.25	6.4	6.6	6.0
100	0.20	0.50	18.6	21.6	18.0
100	0.20	0.75	50.7	63.4	48.3
100	0.30	0.25	6.5	7.0	6.2
100	0.30	0.50	18.6	21.3	15.0
100	0.30	0.75	50.3	58.3	45.9
Mean			23.8	27.1	21.8

payments received, so this is unlikely to be a source of jealousy among the TOs.

4.4. Repeated projects

In this subsection, we consider whether a TO should commit to a shorter task duration in a situation where the same project occurs multiple times, under the CCPM model. As discussed in Section 1, an important motivation for Parkinson's Law is a concern about increased expectations for future performance (Quigley, 2011). To study this issue, we consider an environment with multiple repetitions of a project, where a TO can report a reduced task duration either with or without a commitment, as defined below, to honor that duration for future rounds.

4.4.1. Incentive scheme

We adapt our (α, β, γ) -Scheme under CCPM for use with a project that is repeated multiple times. We consider any milestone $i \in M$. We merge ϕ_j and ψ_j , as defined in (13) and (14), into a single linear function of the reported task duration b'_j with $\theta_j = 1 - \alpha_j = \beta_j - 1$:

$$R_j(b'_j) = 1 + \frac{(b_j + a_j)\theta_j}{b_j - a_j} - \frac{2\theta_j}{b_j - a_j}b'_j, \quad b'_j \in [a_j, b_j], j \in P_i. \quad (19)$$

In reporting his task duration in a single project, as shown in Theorem 5, each TO j reports his “true” duration b_j^* . However, if the same project is repeated multiple times, this may no longer hold true, since the TO may be concerned about receiving lower payments due to a shortened contractual task duration between b_j^* and b_j at future rounds of the project. To address this concern, we provide an option to the TO within his report in any round, say $k \geq 1$, of the same project. That is, while reporting his “true” value b_j^* and obtaining the corresponding incentive payment, the TO may *commit* for future rounds of the project to a duration

$\bar{b}_j \in [b_j^*, b_j^{(k-1)}]$, where $b_j^{(k-1)}$ is the task duration committed to by the TO after the first $k-1$ rounds of the project. By doing so, the TO contractually agrees that this reduction in the upper bound on the task duration is acceptable for all future repetitions of the project. As a special case, the TO makes no new commitment at all in any round k if he simply chooses $\bar{b}_j = b_j^{(k-1)}$.

We provide an incentive scheme that is sufficient to ensure that the payment to each TO j for any report b'_j with *maximum new commitment* is larger than the payment for the same report with any other level of commitment, including no new commitment at all.

(θ, γ) -Scheme

- Fix any milestone $i \in M$. Fix parameters $\theta_j \in (0, 1)$ and $\gamma \in (0, \kappa^0)$ where κ^0 is defined in (9), such that $(1 + \theta_j)^2(1 - \gamma) < 1$ for any $j \in P_i$.
- In round $k \geq 1$ of the project, the realized milestone reward $Z_i^{(k)}$ is shared as follows:
 - Give a fraction $R_j^{(k)}(b'_j)e_j$ of $Z_i^{(k)}$ to any TO $j \in P_i$, where e_j is defined as in (12) and

$$R_j^{(k)}(b'_j) = \left(1 + \theta_j \sum_{t=1}^{k-1} \delta_j^{(t)}\right) R_j(b'_j), \quad b'_j \in [a_j, b_j^{(k-1)}], \quad (20)$$

where $b_j^{(0)} = b_j$ and

$$\delta_j^{(t)} = \frac{b_j^{(t-1)} - b_j^{(t)}}{b_j - a_j} \geq 0, \quad t \geq 1, \quad (21)$$

where $b_j^{(t)} \in [a_j, b_j^{(t-1)}]$ is the task duration committed by the TO up to round t .

- The PM keeps the following fraction of $Z_i^{(k)}$:

$$1 - \sum_{j \in P_i} R_j^{(k)}(b'_j) e_j.$$

End of scheme

We observe that if no reduced task duration is reported as committed in round $k-1$, then $\delta_j^{(k-1)} = 0$, and hence $R_j^{(k)}(\cdot) = R_j^{(k-1)}(\cdot)$. Furthermore, it is clear that we have $\sum_{t=1}^{\infty} \delta_j^{(t)} \leq 1$.

Theorem 8. For projects with multiple identical rounds, the (θ, γ) -Scheme is incentive compatible.

Theorem 8 shows how the use of a greater incentive when a commitment is made incentivizes the TOs to report truthfully. Regarding the allocation of incentive payments under the (θ, γ) -Scheme, we have the following bounds on payments to different parties.

Theorem 9. The (θ, γ) -Scheme is budget balanced. More specifically, in any round k of the project and any fixed milestone $i \in M$, TO j ($j \in P_i$) receives a fraction between $(1 - \theta_j)e_j$ and $(1 + \theta_j)^2 e_j$ of $Z_i^{(k)}$ and the PM receives a fraction between $1 - (1 + \theta_{\max})^2(1 - \gamma)\sigma_i > 0$ and $1 - \theta_{\min}(1 - \gamma)\sigma_i < 1$ of $Z_i^{(k)}$, where $\theta_{\min} = \min_{j \in P_i} \{\theta_j\}$ and $\theta_{\max} = \max_{j \in P_i} \{\theta_j\}$.

Remark 7. Since the (θ, γ) -Scheme is a special case of the (α, β, γ) -Scheme at each individual round, the PM can choose the scheme parameters optimally, as discussed in Section 4.3.

Remark 8. Instead of a single linear function $R_j^{(k)}(b'_j)$, we may alternatively use piecewise linear functions ϕ_j and ψ_j , as in Section 4.1, to distinguish the “additional bonus” for reporting a duration smaller than m_j from the “loss of bonus” for reporting a

Table 3
Improvement in project makespan over the benchmark scheme.

n	OS	Mean	Std. Dev.
25	0.10	10.0	3.5
25	0.20	7.9	3.8
25	0.30	6.9	2.2
50	0.10	7.4	2.2
50	0.20	5.6	2.2
50	0.30	4.2	2.0
100	0.10	8.1	1.9
100	0.20	5.4	2.1
100	0.30	4.5	1.6
Mean		6.7	3.0

duration larger than m_j . Also, as in [Remarks 3](#) and [6](#), we may differentiate the payment across different milestones.

4.4.2. Computational study

In this section, we study the project makespan performance under CCPM planning of our (θ, γ) -Scheme, relative to that of a benchmark scheme, for projects with multiple identical rounds. From [Theorem 8](#), the (θ, γ) -Scheme is incentive compatible, which implies not only that all the TOs report their “true” task durations, but also that they make commitment to the shortened task durations for future rounds. Whereas, in both benchmark Schemes 1 and 2 discussed in [Section 3.4.1](#), the TOs are unwilling to commit to reduced task durations in future, and hence these schemes are vulnerable to Parkinson’s Law. Therefore, we assume as our benchmark that the TOs report their upper bounds on task durations during multiple rounds even after shorter durations are realized. The parameters considered in our data set follow those used in the comparison of our incentive schemes under CPM and CCPM planning in [Sections 4.2](#) and [4.3](#).

[Table 3](#) reports results for the relative percentage improvement in project makespan over the benchmark scheme. The overall mean reduction is 6.7% and the overall standard deviation of the reduction is 3.0%. The results also show that the relative improvement decreases slightly with n , and decreases more noticeably with the OS value.

These results provide the following useful insights to managers. First, a significant improvement in project makespan can be achieved through the use of our incentive scheme. Second, this improvement is more significant in project networks with low density. Third, the variability of improvement between projects is large relative to the mean improvement; hence, it would be useful to identify types of project networks or specific project applications where reduction in makespan can reliably be achieved from our incentive mechanism.

5. Concluding remarks

This paper addresses the pervasive behavioral problem of Parkinson’s Law among employee TOs, which arises at the execution stage of many projects. The effect of Parkinson’s Law is to waste the potential benefit of early task completion. We describe an IC mechanism to resolve Parkinson’s Law for a project planned under CPM. Our mechanism also resolves Student Syndrome. Furthermore, our incentive scheme prevents strategic behavior by any group of TOs. A recent alternative to CPM is CCPM which addresses, but does not fully resolve, Parkinson’s Law. We describe an IC mechanism for a project planned under CCPM. We compare the distributions of incentive payments under the mechanisms for CPM and CCPM, and show that the latter mechanism provides both advantages and disadvantages. Finally, we consider the issue of whether a task operator should commit to earlier task completion for similar future projects.

We discuss insights for practicing project managers. Within both the planning environments of CPM and CCPM, we provide a simple incentive scheme that can easily be followed by TOs within the organization. When this incentive scheme is used, it is in the interests of each TO to report his task(s) as completed, regardless of the actions of other TOs. Hence, if all the TOs act in a self-interested way, Parkinson’s Law is eliminated. Consequently, early task completion times are passed through the project, resulting in improved project performance. Hence, it is possible to eliminate the negative effects of Parkinson’s Law without giving up the detailed control provided by CPM. Moreover, an advantage of using CCPM under our proposed mechanism is greater flexibility in the distribution of typically higher incentive payments to the TOs. This tradeoff informs the management choice between CPM and CCPM for project planning. Finally, the use of appropriate incentives to encourage early completion of tasks should motivate greater efficiencies at the task level. We also provide specific managerial insights following our computational studies in [Sections 3.4.2](#), [4.2](#) and [4.4.2](#). The incentive schemes we propose are easily implementable. We recognize that performance incentives are structured differently by each organization, but our work provides a flexible basis from which various practical schemes can be designed.

Several interesting problems remain open for future research. First, this paper focuses on the timely completion of projects, without the possibility of crashing tasks, which represents an important generalization to consider. Second, the incentive scheme described in [Section 4.4](#) for CCPM projects with multiple rounds should be extended to CPM projects. Third, considering discounting could be a valuable generalization in projects of longer duration. Fourth, IC mechanisms should be developed for other project management control structures, e.g., agile ([Manifesto for Agile Software Development, 2001](#)). Fifth, in practice, our incentive schemes may be extendable to external subcontractors, where there is greater information asymmetry and the possibility of additional strategic behavior. Sixth, our analysis of the long-run implications of commitment to early completion in [Section 4.4](#) should be extended to consider similar tasks that arise in multiple dissimilar projects. Finally, the supervision and motivation of autonomously run tasks is an important topic of current interest ([Linberg, 1999](#); [Wu et al., 2014](#)), for which the implications of our work should be studied. In conclusion, we hope that our work will encourage further research on these important issues.

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Appendix

A.1. Illustration of the (λ, μ) -Scheme

For simplicity, we let $\lambda_{ij} = \lambda_i$ and $\mu_{ij} = \mu_i$ for all tasks j , and $\sigma_i = 1$. We calculate the incentive payments to the TOs

of tasks $j = (2, i)$ and $k = (3, i)$ in Example 1. Note that $P_i = \{(s, 1), (s, 2), (1, 2), (1, 3), (2, 3), (2, i), (3, i)\}$. In Fig. 2, the two numbers in parentheses are the values of r_j and b_j , respectively, and those in brackets are the values of r'_j and b'_j , respectively, for task j . Thus, $Z_i = \rho_i(9 - 8) = \rho_i$. From (4), the slack of task $j = (2, i)$ is $s_j = \mu_i(4 - 3\lambda_i) + (5 - 3\lambda_i) = (4\mu_i + 5) - 3\lambda_i(\mu_i + 1)$, and the slack of task $k = (3, i)$ is $s_k = \mu_i(7 - 6\lambda_i) + (2 - 2\lambda_i) = (7\mu_i + 2) - 2\lambda_i(3\mu_i + 1)$. We also have $D_i = \sum_{\ell} (\mu_i r_{\ell} + b_{\ell}) = 21\mu_i + 20$. We assume that at time $t = 1$, the TOs for tasks $(s, 1)$ and $(s, 2)$ truthfully report that they can reduce their task durations by one time unit. As a result, at the beginning of time $t = 2$, the PM announces earlier accessible times for tasks $(1, 2)$ and $(1, 3)$, which results in reports of a reduced ready time of 2 for each of these tasks.

We consider the incentive payment to TO $j = (2, i)$. At time $t = 3$, after the PM announces an updated accessible time $\xi_h(2) = 3$ for tasks $h \in \{(2, i), (2, 3)\}$, TO j reports an earlier ready time of 3. Therefore, for task j , we have $\tau_j^r = 3$. Assume that, one time unit later at time $t = 4$, the TO for task $(2, i)$ reports that he can reduce the task duration from 5 to 3. Hence, $\tau_j^b = 4$. The early reporting metric for task j is $f_j = (3 + 4)/(2 \times 4 + 5) = 7/13$. Finally, we obtain the incentive payment to TO $j = (2, i)$ for milestone i as follows, from (5):

$$\begin{aligned} & \frac{s_j + D_i \omega (1 - f_j)}{D_i (1 + \omega)} \rho_i \\ &= \frac{(4\mu_i + 5) - 3\lambda_i(\mu_i + 1) + (21\mu_i + 20)(6/13)\omega}{(21\mu_i + 20)(1 + \omega)} \rho_i. \end{aligned}$$

For example, if $\lambda_i = 0.5$, $\mu_i = 1.5$ and $\omega = 1/2$, then the incentive payment to the TO of task $(2, i)$ for its contribution to milestone i is $(29 + 103(6/13))\rho_i/(3(103)) \approx 0.25\rho_i$.

We similarly consider the incentive payment to TO $k = (3, i)$. Assume that, at time 3, the TO for task $(1, 3)$ reports that he can reduce his task duration from 4 to 3. Further assume that, at $t = 5$, one time unit after task $(2, 3)$ starts at 4, the TO reports that he can reduce his task duration from 3 to 2. As a result, at time $t = 6$ when the PM announces an updated accessible time $\xi_{\ell}(5) = 6$ for tasks $\ell \in \{(3, i), (3, t)\}$, a reduced ready time of 6 is reported for task $(3, i)$, hence $\tau_k^r = 6$. Task $(3, i)$ has no reduction in reported duration, hence $b'_k = b_k$ which by definition implies that $r_k^b = T_k = 8$. The early reporting metric for task k is thus $f_k = (6 + 8)/(2 \times 7 + 2) = 7/8$. Finally, we obtain the incentive payment to TO k for milestone i as follows, from (5):

$$\begin{aligned} & \frac{s_k + D_i \omega (1 - f_k)}{D_i (1 + \omega)} \rho_i \\ &= \frac{(7\mu_i + 2) - 2\lambda_i(3\mu_i + 1) + (21\mu_i + 20)(1/8)\omega}{(21\mu_i + 20)(1 + \omega)} \rho_i. \end{aligned}$$

For example, if $\lambda_i = 0.5$, $\mu_i = 1.5$ and $\omega = 1/2$, then the incentive payment to the TO of task $(3, i)$ for its contribution to milestone i is $(28 + 103/8)/(103(3)) \approx 0.13\rho_i$.

A.2. Proof of Theorem 1

Let $D'_i = \sum_{k \in P_i} (\mu_{ik} r'_k + b'_k)$. Since $b'_k \geq a_k$, we have $\kappa_i D_i \leq D'_i \leq D_i$. The fraction of the milestone payment $\sigma_i Z_i$ retained by the PM is the coefficient of $\sigma_i Z_i$ in (6), which is clearly at least $\lambda_i^{\min} D'_i / (D_i (1 + \omega)) \geq \lambda_i^{\min} \kappa_i / (1 + \omega)$ and at most $(\lambda_i^{\max} + \omega) / (1 + \omega)$. Since the PM keeps the remainder of $\sigma_i Z_i$, the incentive scheme is budget balanced.

A.3. Proof of Theorem 2

Consider any milestone $i \in M$ with $Z_i > 0$ and any task $j \in P_i$. From (1) and (3), the milestone payment Z_i is decreasing in both

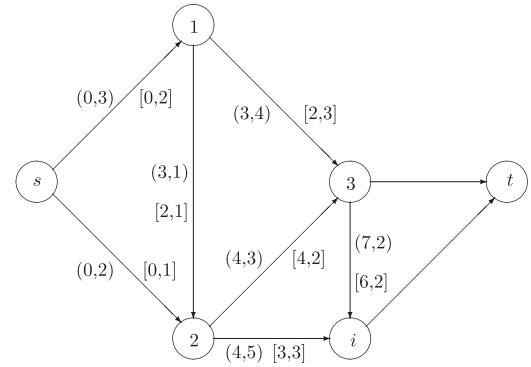


Fig. 2. Computation of task incentives in the (λ, μ) -Scheme.

b'_j and r'_j and in their reporting times τ_j^r and τ_j^b . To interpret this from the perspective of TO j , from (4) and (5), his payment p_{ij} is a function of his reports (r'_j, b'_j) and their reporting times (τ_j^r, τ_j^b) , and is decreasing in any of these variables. Hence, it is clearly beneficial for him to report as early as possible. Now, denote by (r_j^f, b_j^f) and (r_j^f, b_j^f) the “true values” and reported values, respectively, submitted by TO j . Since the TO reports only these two values, only the following scenarios can provide additional payment to TO j based on a false report: (a) $r_j^f < r'_j$; and (b) $b_j^f < b'_j$.

Case (a). Recall that TO j is contractually required to start no later than his reported ready time. Thus, if $r_j^f < r'_j$, then TO j cannot honor the contract, since r'_j is the earliest time at which task j is ready to start.

Case (b). Suppose $b_j^f < b'_j$. Then, due to the contractual requirement in the Certification of Completion, TO j has to start task j earlier than his reported ready time r_j^f , hence

$$r_j^f < r_j^f. \quad (A-1)$$

Let $\delta = r_j^f - r'_j > 0$. The reported false task duration b_j^f cannot be less than $b'_j - \delta$ because of the Certification of Completion requirement, i.e.,

$$b_j^f \geq b'_j - \delta. \quad (A-2)$$

Let s_j denote the slack corresponding to the truthful reports (r'_j, b'_j) and s_j^f the slack corresponding to the false reports (r_j^f, b_j^f) . Then, we have

$$\begin{aligned} s_j^f - s_j &= \mu_{ij} \lambda_{ij} (r'_j - r_j^f) + \lambda_{ij} (b'_j - b_j^f), \quad \text{from (4)} \\ &\leq \lambda_{ij} \delta (1 - \mu_{ij}), \quad \text{from (A-1) and (A-2)} \\ &< 0, \quad \text{from the definitions of } \lambda_{ij}, \delta \text{ and } \mu_{ij}. \end{aligned}$$

Then, from (4) and (5), false reporting by TO j reduces his payment.

A.4. Proof of Corollary 1

The result follows from Theorem 2 and the observation that the total payment that the PM makes to a TO from two independent tasks is the sum of their individual payments.

A.5. Illustration of the weighted (λ, μ) -Scheme

We now provide a numerical example of the Weighted (λ, μ) -Scheme, using Fig. 2. As in Section A.1, we let $\omega = 1/2$ and let $\sigma_i = 1$, $\lambda = \lambda_{ij} = 0.5$ and $\mu = \mu_{ij} = 1.5$ for all tasks j for the fixed milestone i . The weights of arcs $(2, i)$ and $(3, i)$ are both 1. The weight of arc $(2, 3)$ is $1 + 1 = 2$. Hence, the weight of arc $(1, 2)$ is $1 + (1 + 2 + 1) = 5$. We show all the corresponding values (except

Table 4
Example of the Weighted (λ, μ) -scheme.

Task	w_j	r_j	r'_j	b_j	b'_j	$\mu r_j + b_j$	$\lambda(\mu r'_j + b'_j)$	s_j	$w_j s_j$	$y_j^\dagger$	$z_j^\dagger$
(s, 1)	12	0	0	3	2	3	1	2	24	36	12
(s, 2)	5	0	0	2	1	2	0.5	1.5	7.5	10	2.5
(1,2)	5	3	2	1	1	5.5	2	3.5	17.5	27.5	10
(1,3)	2	3	2	4	3	8.5	3	5.5	11	17	6
(2,3)	2	4	4	3	2	9	4	5	10	18	8
(2,i)	1	4	3	5	3	11	3.75	7.25	7.25	11	3.75
(3,i)	1	7	6	2	2	12.5	5.5	7	7	12.5	5.5
Total										132	47.75

$\dagger y_j = w_j(\mu r_j + b_j)$ and $z_j = w_j \lambda(\mu r'_j + b'_j)$.

reporting times) in Table 4. As shown in Section A.1, for tasks $k = (1, 2)$ and $j = (2, i)$, we have $\tau_k^r = 2$, $\tau_k^b = 3$, $\tau_j^r = 3$ and $\tau_j^b = 4$, which give $f_k = (2 + 3)/(2 \times 3 + 1) = 5/7$ and $f_j = 7/13$. Therefore, the TO of task $k = (1, 2)$ receives an incentive payment of $(17.5 + 132(1/2)(2/7))\rho_i/(132(1 + 1/2)) \approx 0.18\rho_i$ towards the completion of milestone i , while the TO of task $j = (2, i)$ gets $(7.25 + 132(1/2)(6/13))\rho_i/(132(1 + 1/2)) \approx 0.19\rho_i$ of the milestone payment. If the same TO owns exactly these two tasks, then his total share of the milestone payment is approximately $0.37\rho_i$.

A.6. Proof of Theorem 3

First, it is in the interest of any TO to report as soon as possible, since p_{ij} is a decreasing function of f_j . Now consider the task(s) assigned to any TO, which form a subnetwork $G' = (N', A')$ of G . Let $A_1 \subseteq A'$ denote the set of root task(s) of the TO, i.e., those tasks that have no predecessor in A' . We consider both aspects of the TO's possible strategy in misreporting r'_j and b'_j for any $j \in A_1$: (a) trading a loss on r'_j for a gain on b'_j ; and (b) trading a loss on task j for gains on tasks $k \in A' \cap V_j$.

- In the Weighted (λ, μ) -Scheme, the relative incentives for reducing r'_j and reducing b'_j remain the same as in the unweighted (λ, μ) -Scheme, as shown in (4). Hence, the TO reports his ready time r'_j truthfully, as shown in the proof of Theorem 2. On the other hand, at duration b'_j , the TO cannot state an earlier-than-actual completion time for task j , because of the Certification of Completion requirement.
- We now show that the TO does not report a later-than-actual completion time for task j , say by Δ time units, in an attempt to trade the immediate loss for a later gain from the extra *unreported* Δ time units of early start for successor tasks $k \in A' \cap V_j$ when it comes to reporting b'_k . Suppose the TO makes such a false report. Then, according to the definition of incentive payments p_{ij} and p_{ik} in (7), the ratio between total possible gained payment and lost payment is at most $(\sum_{k \in A' \cap V_j} w_k \Delta)/(w_j \Delta) < 1$, by construction of the weights. Therefore, both r'_j and b'_j are reported truthfully. We now apply this argument inductively to the root tasks A_{i+1} of $A' \setminus (\cup_{1 \leq t \leq i} A_t)$ for $i \geq 1$ until all tasks in A' have been considered. We conclude that the TO truthfully reports r'_j and b'_j , for $j \in A'$.

A.7. Proof of Corollary 2

Consider any coalition of TOs. Suppose to the contrary that strategic reports of the coalition lead to an increased total payment. Then, when all the tasks owned by members of the coalition are assigned to a single TO, with the same strategic reports, the TO receives more than from truthfully reporting about his individual tasks. This contradicts Theorem 3.

A.8. Vulnerability of Scheme 1

We discuss the possibility that Scheme 1 is vulnerable to false reporting. If a task with maximum duration of b_j is completed in b'_j time units, Scheme 1 pays TO j a bonus of $\bar{\alpha}(b_j - b'_j)$ if $b'_j < b_j$ and requires a penalty of $\max\{\bar{\beta}(b'_j - b_j), \bar{M}_j\}$ if $b'_j > b_j$, where $\bar{M}_j$ is a pre-fixed limit. This scheme works fine for a task that is not a predecessor of another owned by the same TO; early completion earns a profit for a TO, but a delay incurs a penalty. Consequently, each TO reports as soon as he can complete the task. However, Example 2 shows that, when a TO has dependent tasks, he can increase his incentive payments by false reporting.

We now model the decisions of the falsely reporting TO, under the following assumptions.

- We consider a scenario where the duration of only a single task j , which we call the *current* task, is realized at below its worst case estimate, i.e., $b'_j < b_j$, and $b'_k = b_k$ for $k \neq j$.
- There is only a single false reporting TO who may report falsely multiple times. False reporting is worthwhile due to the realized shorter duration of the current task.
- Ready times are ignored, i.e., we assume all tasks are ready as soon as needed. See the first paragraph of Section 4.2 for details.
- Let T denote the set of tasks for which the earliest completion times in the original graph G are decreased as a result of the shortened duration of the current task of the falsely reporting TO, where all other task durations are at their upper bounds. We assume all tasks in T belong to the TO, and form a connected graph $\tilde{G}(T)$.
- At the time when the TO who is falsely reporting knows the actual completion time of the current task, which is earlier than its original worst case estimate, the TO forms his strategy for reporting durations of all tasks in T , based on the updated duration of the current task and the given upper bounds on the durations of all other tasks in T as well as of all the project tasks not in T .

We model the decision problem faced by a TO who may report falsely in order to maximize his incentive payment. We begin with definitions of some parameters used in the formulation. Using the upper bounds $\{b_k\}$ of task durations for all tasks k , both inside T and outside T , the TOs commit earliest completion times for their tasks, which we denote by $\{\bar{E}_k\}$. At the time of forming a reporting strategy, the TO knows the reduced duration of his current task, which results in updated earliest completion times $\{F_j; j \in T\}$, assuming the durations of all other tasks, whether or not in T , are at their given upper bounds $\{b_j\}$. We define three nonnegative variables for each task $j \in T$:

- x_j = reported amount by which the duration of task j is shortened;
- y_j = starting time of task j contractually required by the reports;

z_j = reported completion time of task j , i.e., $z_j = y_j + (b_j - x_j)$.

Solution of the following integer program maximizes the incentive reward received by the TO:

$$\max \sum_{j \in T} x_j$$

subject to: for any $j \in T$,

$$y_j = 0, \quad \text{if } j \text{ is the current task}$$

$$z_j = y_j - x_j + b_j, \quad (\text{by definition of the decision variables})$$

$$y_j = \max \left\{ \max_{\ell \in Q'_j} \bar{F}_\ell, \max_{k \in Q''_j} z_k \right\}, \quad (\text{precedence constraints})$$

$$0 \leq x_j \leq b_j - a_j, \quad (\text{by definition of } x_j)$$

$$F_j \leq z_j \leq \bar{F}_j, \quad (\text{for certification of completion or commitment})$$

where

$$Q'_j = \{\ell \notin T : \ell \text{ is an immediate predecessor of task } j\},$$

$$Q''_j = \{k \in T : k \text{ is an immediate predecessor of task } j\}.$$

We note that the above precedence constraints can be linearized as follows:

$$0 \leq y_j - \bar{F}_\ell \leq \bar{M}(1 - \lambda_\ell^j), \quad \ell \in Q'_j$$

$$0 \leq y_j - z_k \leq \bar{M}(1 - \lambda_k^j), \quad k \in Q''_j$$

$$\sum_{\ell \in Q'_j} \lambda_\ell^j + \sum_{k \in Q''_j} \lambda_k^j \geq 1, \quad \lambda_\ell^j, \lambda_k^j \in \{0, 1\},$$

$$\text{where } \bar{M} = \max_{j \in T} \{\bar{F}_j\}.$$

We now provide an example of our model of false reporting.

Example 3. In Fig. 1, let the set of tasks $\{(s, 1), (1, 2), (1, 3), (2, 3), (2, i), (3, i)\}$ belong to the same TO. Assume that the current job is $(s, 1)$, with an original upper bound duration of 3 and a realized duration of 2. For simplicity, we assume that the lower bound on the duration of any task is 0. The above mixed integer program gives:

$$\text{Maximize } z = x_{s1} + x_{12} + x_{13} + x_{23} + x_{2i} + x_{3i}$$

subject to

$$x_{s1} + z_{s1} = 3,$$

$$x_{12} + z_{12} = z_{s1} + 1,$$

$$x_{13} + z_{13} = z_{s1} + 4,$$

$$x_{23} + z_{23} = z_{12} + 3,$$

$$x_{2i} + z_{2i} = z_{12} + 5,$$

$$x_{3i} + z_{3i} = \max\{z_{13}, z_{23}\} + 2,$$

$$0 \leq x_{s1} \leq 3, \quad 2 \leq z_{s1} \leq 3,$$

$$0 \leq x_{12} \leq 1, \quad 3 \leq z_{12} \leq 4,$$

$$0 \leq x_{13} \leq 4, \quad 6 \leq z_{13} \leq 7,$$

$$0 \leq x_{23} \leq 3, \quad 6 \leq z_{23} \leq 7,$$

$$0 \leq x_{2i} \leq 5, \quad 8 \leq z_{2i} \leq 9,$$

$$0 \leq x_{3i} \leq 2, \quad 8 \leq z_{3i} \leq 9.$$

This model has multiple optimal solutions; for any $0 \leq \theta \leq 1$, an optimal solution is $x_{s1} = 0$, $x_{12} = 0$, $x_{13} = 1 - \theta$, $x_{23} = 1$, $x_{2i} = 1$ and $x_{3i} = \theta$, where $z = 3$. Therefore, although the TO finishes task $(s, 1)$ at time $t = 2$, he can falsely report that the task finishes at time $t = 3$. Then, in private he can start tasks $(1, 2)$, $(1, 3)$ and all his other three tasks one time unit earlier than in the original schedule calculated from upper bound durations. As a result, although the actual duration of each of these five tasks is at its upper bound, the TO, choosing $\theta = 0$, can falsely claim that he uses

Table 5

Computational results for the Weighted (λ, μ) -Scheme with multiple falsely reporting TOs.

n	OS	Imp	Imp-P	Imp-N	Imp-M	PY (%)	MK (%)
50	0.1	990	986	14	374	61.22	2.94
50	0.2	998	996	4	517	75.66	2.53
50	0.3	999	997	3	652	81.81	3.02
100	0.1	999	998	2	423	76.47	2.82
100	0.2	1000	995	5	568	86.88	2.91
100	0.3	999	994	6	622	89.55	2.21
200	0.1	1000	1000	0	170	85.19	2.94
200	0.2	999	999	1	312	91.37	2.52
200	0.3	996	996	4	439	92.44	2.87
Mean		998	996	4	453	82.29	2.75

one time unit less to complete each of tasks $(1, 3)$, $(2, 3)$ and $(2, i)$, without being discovered by the verification of completion report. Thus, the total incentive payment received by the TO is 1 based on honest reporting, and 3 under the false reporting permitted by Scheme 1 in this solution. Hence, the TO has an incentive to report falsely.

A.9. Multiple falsely reporting TOs

We consider multiple falsely reporting TOs, each owning a task of realized duration shorter than its worst-case estimate. Our computational study uses the same setting as in Section 3.4.2, except that each instance is an aggregation of five falsely reporting TOs' subnetworks of tasks and the aggregation is either in parallel or in series, each with probability 0.5. Our computational simulation results are summarized in Table 5, using the same format as in Table 1.

Compared with the results with one single falsely reporting TO (as shown in Table 1), the following changes due to multiple falsely reporting TOs are significant: (a) in nearly all instances, the Weighted (λ, μ) -Scheme improves upon Scheme 1 in terms of reduction either in total incentive payment or in project makespan, or both; (b) the relative payment reduction of the Weighted (λ, μ) -Scheme over Scheme 1 increases by 50%, since more false reporting results in greater incentive payments; and (c) the relative project makespan reduction decreases by 50%, since aggregation of small percentage reductions in subnetwork makespans leads to even smaller percentage reduction of the overall network makespan, especially in the case of parallel subnetworks.

A.10. Proof of Theorem 4

For the TOs, the result is immediate, by construction. It remains to prove the claim about the payment to the PM, which follows from (15) and the following equation:

$$\begin{aligned} \sum_{j \in P_i} e_j &= \frac{\sum_{j \in P_i} (b_j - \tilde{\lambda} m_j)}{B_i} \sigma_i = \frac{B_i - \tilde{\lambda} (A_i + B_i)/2}{B_i} \sigma_i \\ &= \left(1 - \frac{1}{2} \tilde{\lambda} - \frac{1}{2} \kappa^0 \tilde{\lambda}\right) \sigma_i = (1 - \gamma) \sigma_i, \end{aligned}$$

where $A_i = \sum_{j \in P_i} a_j$ and $B_i = \sum_{j \in P_i} b_j$. Therefore, the (α, β, γ) -Scheme is budget balanced.

A.11. Proof of Theorem 5

For each task j , parameters a_j , b_j and m_j are predetermined, and this information, along with P_i , is publicly known and fixed. Therefore, since α_j , β_j and γ are also publicly known and fixed, the only private information that affects the final payments to the TOs is the actual duration of task j , which we denote by b_j^* , for $j \in P_i$.

For each $j \in P_i$, suppose the TO of task j reports a task duration b'_j . If $b'_j < b_j^*$, then the TO does not meet the Certification of Completion requirement. Therefore, we have $b'_j \geq b_j^*$. The incentive payment to TO j is $\phi_j e_j Z_i$ or $\psi_j e_j Z_i$. While e_j is independent of the reported task duration b'_j , all ϕ_j , ψ_j and Z_i are decreasing functions of b'_j . Hence, the incentive payment is reduced if TO j reports a task duration larger than the honest value of b_j^* . Therefore, it is in the interest of TO j to report his honest task duration.

A.12. Proof of Theorem 6

Part (a). Observe that, since $a_j < m_j < b_j$, we have $\alpha_j < 1 < \beta_j$, which satisfies (10). We next prove that $\beta_j(1 - \gamma) < 1$, for $j \in P_i$. We have

$$\begin{aligned} a_j &< b_j \\ (1 + \kappa)a_j &< a_j + \kappa b_j, \quad \text{since } \kappa > 0 \\ \lambda^2((1 + \kappa)a_j)/2 &< \lambda(a_j + \kappa b_j)/2, \quad \text{since } 0 < \lambda < 1 \Rightarrow \lambda^2 < \lambda \\ a_j \gamma \lambda &< a_j \lambda + b_j \gamma - \lambda m_j, \end{aligned}$$

from the definitions of m_j and γ in (16)

$$(b_j - \lambda a_j)(1 - \gamma) < b_j - \lambda m_j, \quad \text{which from (17) satisfies (11).}$$

Part (b). Consider any $j \in P_i$. For the first case, assume $m_j \leq b'_j \leq b_j$. Denote by z_j^{CP} and z_j^{CC} the payments to TO j as a fraction of Z_i under the (λ, μ) -Scheme for the CPM system with $\mu = 0$ and the (α, β, γ) -Scheme for the CCPM system, respectively. Then, from (4) and (5) with all $\mu_{ik} = 0$, and from (12) and (13) with $\tilde{\lambda} = \lambda$, we have

$$\begin{aligned} z_j^{CP} &= \frac{b_j - \lambda b'_j}{\sum_{k \in P_i} b_k} \sigma_i, \\ z_j^{CC} &= \frac{(b_j - \alpha_j m_j) - (1 - \alpha_j) b'_j}{b_j - m_j} \left(\frac{b_j - \lambda m_j}{\sum_{k \in P_i} b_k} \right) \sigma_i. \end{aligned}$$

Note that if $b'_j = m_j$, we have $z_j^{CP} = z_j^{CC}$, where both are equal to $(b_j - \lambda m_j) \sigma_i / \sum_{k \in P_i} b_k$. On the other hand, if $b'_j = b_j$, we get $z_j^{CP} = z_j^{CC}$, where both are equal to $b_j(1 - \lambda) \sigma_i / \sum_{k \in P_i} b_k$ according to the settings in (17). Therefore, z_j^{CP} and z_j^{CC} are identical linear functions of b'_j . The proof of the other case where $a_j \leq b'_j \leq m_j$ is similar.

The result that the PM receives the same payment in the two incentive schemes follows directly from the similar result for every TO, and from budget balance.

A.13. Proof of Theorem 7

From (13) and (14), payment to any TO j is nondecreasing in both α_j and β_j . Hence, the first part of conclusion (b) in Theorem 6 together with the setting defined by (18) implies the first part of the theorem.

From Theorem 4, the fractions of Z_i that the PM receives under the two schemes are both in the same interval $[1 - \beta(1 - \gamma) \sigma_i, 1 - \alpha(1 - \gamma) \sigma_i]$.

A.14. Proof of Theorem 8

From (20), we have $R_j^{(1)}(b'_j) = R_j(b'_j)$, and $R_j^{(k)}(b'_j) = R_j^{(k-1)}(b'_j) + \theta_j \delta_j^{(k-1)} R_j(b'_j)$, for any $k \geq 2$. That is, in any subsequent round k of the project, the fraction of $Z_i^{(k)}$ given to TO j is $\theta_j \delta_j^{(k-1)} R_j(\cdot) e_j$ more than that for the previous round of the project. Therefore, if TO j commits to a reduced duration $b_j^{(k-1)}$ to the greatest extent possible in the first $k - 1$ rounds, he maximizes

his payment fraction from the next round on. Moreover, from the additivity of $\{\delta_j^{(t)}\}$ in the payment fraction, such commitments from TO j are made as soon as possible.

A.15. Proof of Theorem 9

It follows from (19) that $1 - \theta_j \leq R_j(\cdot) \leq 1 + \theta_j$. Hence, at any round k , since $\sum_{t=1}^{\infty} \delta_j^{(t)} \leq 1$, the payment fraction $R_j^{(k)}(\cdot)$ of $Z_i^{(k)}$ to TO j satisfies the following:

$$1 - \theta_j \leq R_j(\cdot) \leq R_j^{(k)}(\cdot) \leq (1 + \theta_j) R_j(\cdot) \leq (1 + \theta_j)^2.$$

The bounds for the payment to the PM follow from the corresponding proof for the second part of Theorem 4. Since by construction all the available milestone payment is distributed to the TOs and the PM, the payment scheme is budget balanced.

References

- Abu-Hilje, S. F., & Ibbs, C. W. (1989). Schedule-based construction incentives. *Journal of Construction Engineering and Management*, 115, 430–443.
- Alberini, A., Longo, A., Tonin, S., Trombetta, F., & Turvani, M. (2005). The role of liability, regulation and economic incentives in brownfield remediation and redevelopment: Evidence from surveys of developers. *Regional Science & Urban Economics*, 35, 327–351.
- Arditi, D., & Yasamis, F. (1998). Incentive/disincentive contracts: Perceptions of owners and contractors. *Journal of Construction Engineering and Management*, 124, 361–373.
- Bayiz, M., & Corbett, C. J. (2005). Coordination and incentive contracts in project management under asymmetric information. In *Anderson Graduate School of Management*. Working paper.
- Bower, D., Ashby, G., Gerald, K., & Smyk, W. (2002). Incentive mechanisms for project success. *Journal of Management in Engineering*, 37–43.
- Bubshait, A. A. (2003). Incentive/disincentive contracts and its effects on industrial projects. *International Journal of Project Management*, 21, 63–70.
- Bukhari, S., & Malik, A. A. (2012). Determining the factors affecting the accuracy of effort estimates for different application and task types. *10th International Conference on Frontiers of Information Technology*, 41–45.
- Chen, T., Klastorin, T., & Wagner, M. R. (2015). Incentive contracts in serial stochastic projects. *Manufacturing & Service Operations Management*, 17(3), 290–301.
- Demeulemeester, E., Vanhoucke, M., & Herroelen, W. (2003). Rangen: A random network generator for activity-on-the-node networks. *Journal of Scheduling*, 6(1), 17–38.
- Enlume (2018). Discover product trends and delight customers. <http://www.enlume.com/industries/retail/> 2019.
- Goldratt, E. M. (1997). *The critical chain*. Great Barrington, MA: North River Press.
- Golenko-Ginzburg, D. (1988). On the distribution of activity time in PERT. *Journal of the Operational Research Society*, 39, 767–771.
- Growwire.com (2019). When the market called for shorter product life cycles, these companies hatched an answer. <https://www.growwire.com/product-life-cycle2019>.
- Gutierrez, G. J., & Kouvelis, P. (1991). Parkinson's law and its implications for project management. *Management Science*, 37, 990–1001.
- Hall, N. G. (2012). Project management: Recent developments and research opportunities. *Journal of Systems Science and Systems Engineering*, 21, 129–143.
- Hall, N. G. (2013). Project management: Further research opportunities. In C. Schwindt, & J. Zimmermann (Eds.), *Handbook of project management and scheduling*, 2, 945–970.
- Hall, N. G. (2016). Research and teaching opportunities in project management. *Tutorials in Operations Research*. <https://doi.org/10.1287/educ.2016.0146>, 2019. INFORMS, Catonsville, MD.
- Herroelen, W., & Leus, R. (2001). On the merits and pitfalls of critical chain scheduling. *Journal of Operations Management*, 19, 559–577.
- Herten, H. J., & Peeters, W. A. R. (1986). Incentive contracting as a project management tool. *Project Management*, 4, 34–39.
- Hill, J., Thomas, L. C., & Allen, D. E. (2000). Experts' estimates of task durations in software development projects. *International Journal of Project Management*, 18, 13–21.
- Hu, X., Cui, N., & Demeulemeester, E. (2015). Effective expediting to improve project due date and cost performance through buffer management. *International Journal of Production Research*, 53(5), 1460–1471.
- Hurwicz, L. (1960). Optimality and informational efficiency in resource allocation processes. In *Mathematical methods in the social sciences*, ed. Arrow, Karlin and Suppes. Stanford University Press.
- Ibbs, C. W. (1991). Innovative contract features for construction. *Construction Management and Economics*, 9, 157–169.
- Kerzner, H. (2009). *Project management: A systems approach to planning*. In *Scheduling and controlling*. Hoboken, NJ: 10th ed. Wiley.
- Klastorin, T. (2012). *Project management: Tools and tradeoffs* (3rd). NJ, USA: Pearson, Upper Saddle River.

- Kog, Y. C., Chua, D. K. H., Loh, P. K., & Jaselskis, E. J. (1999). Key determinants for construction schedule performance. *International Journal of Project Management*, 17, 351–359.
- Krakowski, M. (1974). PERT and Parkinson's law. *Interfaces*, 5, 35–40.
- Kwon, H. D., Lippman, S. A., McCardle, K. F., & Tang, C. S. (2012). Project management contracts with delayed payments. *Manufacturing & Service Operations Management*, 12(4), 692–707.
- Kwon, H. D., Lippman, S. A., & Tang, C. S. (2010). Optimal time-based and cost-based coordinated project contracts with unobservable work rates. *International Journal of Production Economics*, 126, 247–254.
- Leach, L. P. (1999). Critical chain project management improves project performance. *Project Management Journal*, 30, 39–51.
- Linberg, K. R. (1999). Software developer perceptions about software project failure: A case study. *Journal of Systems and Software*, 49, 177–192.
- Manifesto for Agile Software Development (2001). <http://agilemanifesto.org> 2019.
- Margraf, B. (2018). Six methods for the estimation of activity duration in project management. <https://smallbusiness.chron.com/six-methods-estimation-activity-duration-project-management-41782.html>, 2019.
- Myerson, R. (1979). Incentive compatibility and the bargaining problem. *Econometrica*, 47, 61–73.
- Myerson, R. (1982). Optimal coordination mechanisms in generalized principal-agent problems. *Journal of Mathematical Economics*, 11, 67–81.
- Myerson, R. (1986). Multistage games with communications. *Econometrica*, 54, 323–358.
- Parkinson, C. N. (1955). Parkinson's law. *Economist*.
- Parkinson, C. N. (1958). *Parkinson's law: The pursuit of progress*. London: John Murray.
- Quigley, J. (2011). Personal communication.
- Radulescu, C., & Heales, J. (2005). Incentives and their effects on information systems projects. *Proceedings of the 13th European Conference on Information Systems, Regensburg, Germany*.
- Raz, T., Barnes, R., & Dvir, D. (2003). A critical look at critical chain project management. *Project Management Journal*, 36, 24–32.
- Schonberger, R. J. (1981). Why projects are “always” late: A rationale based on manual simulation of a PERT/CPM network. *Interfaces*, 11, 66–70.
- Sommer, S. C., & Loch, C. H. (2009). Incentive contracts in projects with unforeseeable uncertainty. *Production and Operations Management*, 18, 185–196.
- Steyn, H. (2001). An investigation into the fundamentals of critical chain project scheduling. *International Journal of Project Management*, 19(6), 363–369.
- The Royal Swedish Academy of Sciences (2007). Mechanism design theory. In *Scientific Background on the Sveriges Riksbank Prize in Economic Sciences in Memory of Alfred Nobel 2007*.
- Vanhoucke, M., Coelho, J., & Batselier, J. (2016). An overview of project data for integrated project management and control. *Journal of Modern Project Management*, 3(2), 6–21.
- Williams, T. M. (1995). What are PERT estimates? *Journal of the Operational Research Society*, 46, 1498–1504.
- Wu, Y., Krishnan, V., & Ramachandran, K. (2014). Managing cost salience and procrastination in projects: Compensation and team composition. *Production and Operations Management*, 23(8), 1299–1311.
- Wysocki, R. K. (2009). *Effective project management*. IN, USA: 5/e. Wiley, Indianapolis.